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Multilayer interdependencies in the banking system of Japan: correlation dynamics and determinants

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In this paper, we analyse the multilayer architecture of the correlation dynamics among Japanese banks over the period from 1982 to 2011. This period encompasses nearly all significant events in the Japanese economy over the past few decades. The two layers we consider represent the loan portfolio correlations stemming from the credit market and the stock return correlations arising from the equity market, respectively. We find that in these different market layers, the correlations can have different structures and they may imply different signals of systematic and non-systematic risk components in the banking system. In addition, our results show that in spite of a certain degree of persistence, the interactions among banks in each layer also evolve over time. Furthermore, utilising the characteristics and the balance sheet information of individual banks, we find that the similarities in this kind of fundamental information can only partially explain the formation of correlations in each layer. In addition, overall, no explanatory variables based on quantitative or qualitative similarities can predict future correlations better than contemporaneous ones.

Keywords: Loan correlations; Stock return correlations; Systemic risk; Principal component analysis; Correlation filtered methods; Determinants of correlations

1. Introduction

Portfolio similarities and correlations among financial institutions (in the following banks for brevity) have been widely seen as one of the major factors that would jeopardise the stability of the financial system. In particular, the more asset portfolios of banks are aligned or correlated, the more they are jointly exposed to systemic risk in the underlying assets. This is mainly because, first, if banks have highly similar holdings, even a minor negative shock to a single bank can spill over to the others via asset fire sales. More specifically, if the shock forces a bank to sell its assets, such selling could create downward pressure on the market prices of the held assets, thereby affecting other banks that hold similar assets. This process continues to repeat if in turn the other affected banks are also forced to sell their assets in order

to shore up their balance sheets. Hence, similar asset portfolios pave a potential pathway toward the instability of the financial system in the sense that they allow fire sales to propagate local shocks across the balance sheets of different banks (see Cifuentes *et al.* 2005, May and Arinaminpathy 2010, Huang *et al.* 2013, Caccioli *et al.* 2015, Greenwood *et al.* 2015, Barucca *et al.* 2021, among others). Second, in the presence of similar portfolios, a shock to a common asset class may simultaneously affect many banks, thus, increasing the likelihood of systemic crises (e.g. Acharya and Yorulmazer 2007, Wagner 2008, 2010).

It is also worthwhile to emphasise that although each individual bank can optimally have a diversified portfolio that helps to reduce the risk (for instance by allocating the holdings into different asset classes or different investments in various economic sectors), at the aggregate level, if banks follow similar diversification strategies, individually optimal strategies will undesirably increase the systemic risk in the

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whole system (e.g. see Wagner 2008, 2010). Furthermore, when the number of banks affected by shocks is sufficiently high, it may constrain the set of possible policy options for regulators (Acharya and Yorulmazer 2007): They may have no other choices but to bail out the affected banks since they cannot let many banks simultaneously fail. Nevertheless, such a ‘too-many-to-fail’ guarantee may then induce banks to ex-ante have jointly correlated lines of business or similar investment strategies such that they can increase the likelihood of being rescued when things go wrong. This will eventually result in severe financial crises that have a substantial impact on the whole economy because of a high degree of similarity among banks.

In a similar vein, various studies also suggest that loan portfolio overlaps in the credit market could give rise to the propagation of financial distress inside the banking system as well as between this system and the other spheres of the economy (e.g. see Fujiwara *et al.* 2009, Hellwig 2009, Lux 2016, Ramadiah *et al.* 2020). For example, on the one side, if banks lend to the same firms or to different firms but of the same sector (i.e. banks are at a high level of joint exposures via common debtors), a shock to these firms or sectors may have a pervasive impact on the whole banking system. On the other side, when non-financial corporations or industries rely on only a few common creditor banks to finance themselves, a reduction in their credit supply may subsequently lead to a severe effect on the real sector and the whole economy.[†] Indeed, a non-random and hierarchical network structure of bank-sector or bank-firm credit relationships has been often observed in different credit markets worldwide (e.g. see Fujiwara *et al.* 2009, Fricke and Roukny 2020, Luu and Lux 2018, 2019). Accordingly, many banks do not select their borrowers randomly but prefer candidates with certain characteristics (or those that fulfill certain credit policy conditions) more than others, which in turn leads to overlaps in their lending portfolios. In another study, Luu (2021) examines the loan portfolio correlations among banks concerning credit provision to large firms in Japan. The findings indicate that distinct states within the Japanese credit market can be delineated by observing the evolution of the most robust significant principal components.

From another perspective, the understanding of correlations between banks’ assets and equity prices can provide important implications for both risk management at the micro level and systemic risk supervision at the macro level. First, the gains from diversification, which can aid an individual bank in mitigating or eradicating portfolio risk, are heavily contingent upon the correlations among its assets, investments, or defaultable loans (e.g. see Markowitz 1952, Tobin 1958, Sharpe 1964, Lintner 1965a, 1965b). Related to this, it should be worthwhile to emphasise that the understanding of systematic risk and non-systematic risk factors driving the interdependencies among financial assets has also been a central issue of the finance literature (e.g. see also Rosenberg and

Marathe 1976, Ross 1976, Chen *et al.* 1986, Fama and French 1992, 1993). In general, a systematic risk factor is often referred to as one that arises from exposure to the whole market, which therefore cannot be diversified away. In contrast, non-systematic risk drivers are those that only affect a subset of the market and, thus, are diversifiable. Second, moving onto the perspective of regulators, the degree of correlations among banks is also an important indicator of systemic risk of the whole banking system (e.g. see Das and Uppal 2004, Lehar 2005, Haldane *et al.* 2007, Kritzman *et al.* 2011, Billio *et al.* 2012, Luu 2021). Specifically, a higher level of correlations indicates that banks are more unified or tightly coupled together. Under such a market structure, an adverse shock can propagate widely and have a profound effect on the whole system. Furthermore, it has been widely observed that correlations are time-varying and often display strongly asymmetric dynamics, i.e. they are much greater after a negative shock than after a positive shock of the same magnitude (e.g. Ang and Chen 2002, Barndorff-Nielsen and Shephard 2004, Cappiello *et al.* 2006, Billio *et al.* 2012). In relation to the macroeconomic conditions, the economy typically shifts between a state of economic growth associated with low financial volatility and a state in which we observe both economic contraction as well as a highly turbulent financial market (IMF 2009, Billio *et al.* 2012). Hence, understanding the level and dynamics of correlations is crucial for identifying various critical periods in both the market and the economy.

In summary, the aforementioned discussion of related literature shows that systemic failures may not only originate from direct credit linkages[‡] alone but may also arise from various indirect interdependencies among banks, e.g. via overlapping portfolios, correlated business lines, or correlations in their equity prices. Indeed, it has been suggested that under certain circumstances, these indirect relationships may have a greater contagion potential than interbank credit (e.g. see Clerc *et al.* 2016, Barnett *et al.* 2022, among others). Nevertheless, despite the importance of a more comprehensive consideration of the different propagation channels together, the relevant literature on the multilayer architecture of relationships among banks has been relatively sparse (Battiston and Martinez-Jaramillo 2018). While the empirical research has concentrated almost entirely on single layers of interdependencies, only a few studies have recently attempted to consider the multilayer architecture of the banking system and jointly analyse the behaviours and interactions of its different layers.

For example, Poledna *et al.* (2015) introduce an approach to quantify systemic risk in the multi-layer network that captures four different types of financial exposures among banks in the banking system of Mexico: Derivatives, securities, foreign exchange, and deposits & loans. The authors show

[†] Various financial as well as real effects of the credit crunch have been well documented in the literature, for example, see Khwaja and Mian (2008), Ongena *et al.* (2015), Jorda *et al.* (2013), Bassett *et al.* (2014), Cingano *et al.* (2016), Gambetti and Musso (2017), Bentolila *et al.* (2018), Amiti and Weinstein (2018), and Alfaro *et al.* (2021).

[‡] For example, see Allen and Gale (2000), Nier *et al.* (2007), Imakubo and Soejima (2010), Haldane and May (2011), Gorton and Metrick (2012), Acemoglu *et al.* (2015), and Luu *et al.* (2021) for further discussions on different types of direct linkages among banks by which financial distress can propagate across the whole system. Refer to Glasserman and Young (2016) for a more comprehensive review on various types of network interconnections as potential channels of contagion.

that it is crucial to consider the multilayer network architecture of the interactions among banks, and that separately analysing individual layers will drastically underestimate systemic risk. Brunetti *et al.* (2019) study a two-layer network, in which the first layer represents the correlations of bank stock returns (namely the correlation network), while the second layer captures interbank lending transactions (namely the physical network). They find that prior to the global financial crisis, the two layers behave similarly. Nevertheless, during the crisis, the former becomes denser while the latter highlights a marked decrease in interconnectedness. Furthermore, these two layers behave differently in response to monetary and macroeconomic shocks: The correlation-based layer forecasts financial crises, while the interbank lending layer signals liquidity problems. From another aspect of indirect interdependencies, Luu and Lux (2019) study the joint exposures among banks in lending to non-financial firms in different layers representing different industrial sectors of the Spanish economy. They find that the multilayer network of the joint exposures among banks exhibits a very heterogeneous structure of interdependencies among layers: On the one hand, the lending portfolios of banks to a number of borrower sectors are highly similar; on the other hand, little or no overlaps are observed in many pairs of other sectors. Compared to various null network models, they find that the two banking groups, namely the commercial banks and savings banks, exhibit a very homogeneous sectoral composition in their loan portfolios.

Over the last few decades, there has been a growing literature investigating the role of interconnections among different economies, markets, sectors, and financial and non-financial institutions. To this end, various econometric and statistical approaches have been proposed and applied to infer the network linkages and to identify potential channels for contagion and amplification of shocks (e.g. see Billio *et al.* 2012, Diebold and Yilmaz 2014, Diebold and Yilmaz 2015, Barigozzi and Hallin 2017, Wang *et al.* 2018, Akhtaruzzaman *et al.* 2021). Network science also provides useful tools for extracting important interrelation patterns from the raw correlation matrices of financial data. Examples of contributions include, among others, Mantegna (1999), Tumminello *et al.* (2005), Di Matteo *et al.* (2010), Aste *et al.* (2010), Massara *et al.* (2016) and MacMahon and Garlaschelli (2015), in which different network algorithms such as Minimum Spanning Tree (MST) and Planar Maximally Filtered Graph (PMFG), or community detection methods have been used to analyse the local, mesoscale and global interactions among different financial assets or listed companies.[†]

Recently, the multiple relations among financial assets have been jointly analysed in several studies. For instance, Wang *et al.* (2020) study the multiple interdependencies among 24 Chinese publicly-listed financial institutions from three separate layers: The mean (return) spillover, the volatility spillover and the extreme risk spillover have been extracted from Granger causality tests of these three indicators. Accordingly, each of these layers has been found to have a unique

structure and dynamics, and the understanding of different behaviours of different layers can shed light on the assessment and management of systemic risk in the system. Billio *et al.* (2021) proposed a Bayesian semiparametric model for temporal networks of two layers where both intralayer and interlayer connectivities are present. The first layer captures return linkages, while the second one represents the network of volatilities. Furthermore, interlayer linkages, i.e. from asset returns to volatilities and vice versa, indicate risk premium linkages, and leverage linkages among the financial assets, respectively. The proposed method was applied to analyse a European financial network of more than 400 firms based in Germany, France, and Italy before and after the spread of COVID-19. The authors showed that the COVID-19 pandemic has had a heterogeneous impact on different layers. In particular, it has increased the volatility and risk premium linkages but reduced the leverage linkages among the considered financial assets.

In our present work, we extend the existing literature on the multilayer nature of the interdependencies among banks in various directions. First, we jointly analyse loan portfolio correlations stemming from the credit market and stock return correlations arising from the equity market in Japan. These two market layers, hence, capture two distinct sources of coupling behaviours and contagion channels among Japanese banks. Second, using a relatively long data sample (from 1982 to 2011), we aim to analyse and compare the structures and dynamics of these two correlation layers over the years and examine how they respond to different shocks. Furthermore, we decompose the correlations into different groups of risk factors and investigate the importance of the characteristics and the balance sheet information of banks in explaining the emergence of such correlations.

Among others, we observe that in the equity and credit market layers, the correlations can have different structures and temporal dynamics. In addition, although there is a certain degree of correlation persistence across several consecutive years, the interactions among banks in each market also evolve over time with different groups of banks dominating in different phases of temporal dynamics. Moreover, we find that the correlations in each market layer are mainly governed by a global (common) factor with a more pervasive impact in the system and a couple of non-systematic factors that only influence some groups of banks. After filtering out the effects of the common factor in each layer, we further explore the determinants of non-systematic correlations using the characteristics and the balance sheet information of banks. Our results show that the similarities in the fundamental information and the characteristics of banks can only partially explain the residual correlations. On top of that, overall, no other explanatory variables can predict future correlations better than contemporaneous correlations.

Our conjecture is that there may exist other factors that are also responsible for the emergence of co-movements in the two market layers, especially those from the demand side. For example, investors in the equity market layer may select stocks and adjust their portfolios based on a certain number of other criteria, which in turn also leads to correlations among the stock returns of these banks. Examples of these criteria include investment categories, preferred habitat, information

[†] See Marti *et al.* (2021) for a more comprehensive review of networks inferred from financial assets.

diffusion due to some market friction, and similar investment sentiment (e.g. see Barberis and Shleifer 2003, Barberis *et al.* 2005, among others). Loan correlations stemming from the credit market layer may also depend on other primitive factors that affect the matching and maintenance of the credit relationships between banks (lenders) and firms (borrowers). There is ample evidence suggesting that screening and credit rationing are also based on the characteristics as well as the fundamental information of borrowers or on how close the credit relationships between lenders and borrowers are (e.g. see Ferri and Murro 2015, Ono *et al.* 2015, Ruan *et al.* 2018, Luu and Lux 2019, Anderson *et al.* 2022). A certain level of persistence of loan correlations over the years may also originate from the fact that borrower firms do not often change their main lenders during a short time period (e.g. Ono *et al.* 2015).

The major take-home message of our paper is that loan portfolio overlaps of banks constitute an important layer of connectivity besides stock return correlations and interbank lending. We find that the correlations in loan portfolios are characterised by structural features and dynamic evolution that differ from those of stock return correlations. Hence, comovements in the stock market are not simply the consequence of similar lending policies, but are mostly driven by other factors. While the network structures of both the stock market layer and the loan portfolio layer evolve over time, the hubs we find in both layers at any point in time are typically different. This underlines that loan synchronisation should be seen as an additional source of financial risk and contagion that is not already encapsulated in other network layers.

The rest of this paper is organised as follows: In Section 2, we briefly describe the data and methods used in our study. Section 3 summarises our main findings for loan portfolio correlations stemming from the credit market and stock return correlations originating from the equity market. After that, in Section 4 we examine the role of banks' characteristics and balance sheet information in explaining such correlations. Finally, further discussions and concluding remarks are in Section 5.

2. Data and methods

2.1. Data

To investigate inter-dependencies among Japanese banks in different financial markets, we use two separate data sets: (i) the first data set provides the daily stock prices of listed Japanese banks in the equity market and (ii) the second data set contains yearly credit linkages between banks and large firms in the credit market of Japan during the period from 1982 to 2011. The analysed period encompasses nearly all significant market states in the Japanese economy over the past few decades, including the asset bubble period, the bubble crash, the subsequent lost decade, and the aftermath of the global financial crisis. While the former data set was extracted from Thomson Reuters Datastream and Yahoo Finance, the latter data set was obtained from firms' financial statements

and the survey by Nikkei Media Marketing, Inc. in Tokyo (commercially available). The two datasets are then merged based on banks' unique identifiers.

From the daily stock prices, we use moving windows to extract the yearly correlations among banks stemming from the equity market. In contrast, with the information on yearly credit relationships from banks to firms, we can compute the yearly correlations among banks in lending to firms (see also Luu 2021). The methods used to extract and analyse correlations are summarised in section 2.2.

2.2. Methods

2.2.1. Stock return correlations. We shall now start with the first layer of the correlations among banks, i.e. the correlations arising from their stock returns in the equity market. Assuming that $P_{i,t}$ represents the stock price of bank i at time (date) t within a (yearly) time window $[T_w, T_{w+1}]$ of length T (days), the log-return associated with this bank is defined as

$$R_{i,t} = \ln(P_{i,t+1}) - \ln(P_{i,t}), \quad (1)$$

for $i = 1, 2, \dots, N_B$, and N_B is the number of banks.

We then obtain the standardised stock return $r_{i,t}$ of bank i at time t as

$$r_{i,t} = \frac{R_{i,t} - \langle R_{i,t} \rangle}{\sigma[R_{i,t}]}, \quad (2)$$

where $\langle R_{i,t} \rangle$ and $\sigma[R_{i,t}]$ are the average and the standard deviation of $R_{i,t}$ over the considered time period $[T_w, T_{w+1}]$, respectively.

Over each period $[T_w, T_{w+1}]$, the cross-correlation matrix of N_B stock returns of banks is defined as

$$C_{Stock} = \{C_{ij,Stock}\}_{N_B \times N_B} = \frac{1}{T} \mathcal{R} \mathcal{R}^{Tr}, \quad (3)$$

with $\mathcal{R} = \{r_{i,t}\}_{N_B \times T}$ being a standardised return matrix of size $N_B \times T$.[†] Note that the estimated correlation matrix C_{Stock} is non-singular if T is larger than N_B (e.g. see Bodnar *et al.* 2014).

2.2.2. Loan portfolio correlations. In the following, we will describe the procedure used to derive the second layer of correlations between banks arising from their lending activities in the credit market. To this end, let us assume that over a certain period (e.g. within a particular year τ), a network of bank-firm credit linkages is characterised by a weighted matrix $W^{b-f}(\tau) = \{w_{ij}^{b-f}\}_{N_B \times N_F}$, where $i = 1, 2, \dots, N_B$ standing for N_B banks, and $j = 1, 2, \dots, N_F$ representing N_F firms (Marotta *et al.* 2015, Luu 2021). Each element w_{ij}^{b-f} indicates the aggregate monetary amount of loans that bank i lends to firm j over the year τ , and each row vector $W_i^{b-f} = [w_{i1}^{b-f}, w_{i2}^{b-f}, \dots, w_{iN_F}^{b-f}]$, hence, implies the loan portfolio of bank i over different borrower firms.

[†] Throughout this paper, for any matrix or vector X , the notation X^{Tr} stands for the transposition of X .

Following Luu (2021), in the first step we define the standardised loan L_{ij}^{b-f} from bank i to firm j as

$$L_{ij}^{b-f} = \frac{w_{ij}^{b-f} - \langle w_{ij}^{b-f} \rangle}{\sigma_{ij}^w}. \quad (4)$$

In equation (4), for each bank i , the two quantities $\langle w_{ij}^{b-f} \rangle$ and σ_{ij}^w represent the average and the standard deviation of w_{ij}^{b-f} over all firms $j = 1, 2, \dots, N_F$, respectively. In the next step, given the standardised loan matrix $\mathcal{L}^{b-f} = \{L_{ij}^{b-f}\}_{N_B \times N_F}$, we can compute the cross-correlation matrix of loan portfolios as

$$C_{Loan} = \{C_{ij,Loan}\}_{N_B \times N_B} = \frac{1}{N_F} \mathcal{L} \mathcal{L}^{Tr}. \quad (5)$$

Notice that each element $C_{ij,Loan}$ indicates the overall correlation coefficient of the loan portfolios of bank i and bank j . A positive (negative) value of $C_{ij,Loan}$ indicates that the lending amounts provided by the two banks over different firms are positively (negatively) correlated.†

2.2.3. Correlation analysis. Identification of Significant Components:

We now describe a unified procedure used for the analysis of a given correlation matrix C . Let us, in general, assume that

$$C = \frac{1}{K} \mathcal{X} \mathcal{X}^{Tr}, \quad (6)$$

with $\mathcal{X} = \{X_{ij}\}_{N \times K}$ is a standardised matrix (i.e. $\langle X_{ij} \rangle = 0$ and $\text{Var}[X_{ij}] = 1$) with the size $N \times K$ ($K > N$). Note that relating to our present work, the correlation matrix C can be either $C = C_{Stock} = \frac{1}{T} \mathcal{R} \mathcal{R}^{Tr}$ defined in equation (3) or $C = C_{Loan} = \frac{1}{N_F} \mathcal{L} \mathcal{L}^{Tr}$ as in equation (5), and N is, thus, the number of banks N_B .

Our first main object is to quantify how *statistically* significant the correlations among banks are with respect to a suitable random benchmark. To this end, the distribution of the eigenvalues of the empirical correlation matrix will be compared with that of a hypothesised correlation matrix $C^{hyp.}$ computed from the chosen random null model. Denote $\{\lambda_i^{hyp.}\}_{i=1}^N$ the eigenvalues of $C^{hyp.}$ and their density function is $\text{pdf}(\lambda^{hyp.})$. According to the Marchenko-Pastur law in Random Matrix Theory (RMT) (e.g. Marčenko and Pastur 1967, Tao and Vu 2012), if the elements X_{ij} of $\mathcal{X} \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$, then in the limit $N, K \rightarrow \infty$ with $Q = \frac{K}{N} \rightarrow a$ (a is constant and $a > 1$), the theoretical distribution $\text{pdf}(\lambda^{hyp.})$ of the eigenvalues is given by

$$\begin{aligned} & \text{pdf}(\lambda^{hyp.}) \\ &= \begin{cases} \frac{Q}{2\pi\sigma^2} \frac{\sqrt{(\lambda_{\max}^{RMT} - \lambda^{hyp.})(\lambda^{hyp.} - \lambda_{\min}^{RMT})}}{\lambda^{hyp.}} & \text{if } \lambda_{\min}^{RMT} \leq \lambda^{hyp.} \leq \lambda_{\max}^{RMT}, \\ 0 & \text{elsewhere.} \end{cases} \end{aligned} \quad (7)$$

† Similarly, C_{Loan} is non-singular if $N_F > N_B$, which is often the case in bank-firm credit networks.

In equation (7), $\lambda_{\max}^{RMT}$ and $\lambda_{\min}^{RMT}$ are the upper bound and the lower bound of eigenvalues predicted by RMT, respectively.‡ In particular,

$$\lambda_{\max}^{RMT} = (1 + \sqrt{1/Q})^2 \quad \text{and} \quad \lambda_{\min}^{RMT} = (1 - \sqrt{1/Q})^2. \quad (8)$$

By comparing the spectrum of the empirical correlation matrix C with the theoretical one mentioned in equation (7), we can examine by how much the structure of empirical correlations deviates from a system of pure noise. More specifically, the empirical eigenvalues that belong to the interval $[\lambda_{\min}^{RMT}, \lambda_{\max}^{RMT}]$ are often regarded as the random bulk that provides no genuine information, and only those that are beyond such an interval will carry significant signals of the correlations between banks (e.g. see Plerou *et al.* 2000, 2002, Zheng *et al.* 2012, Meng *et al.* 2014, Lux *et al.* 2020, Luu 2021, among others).

Correlation Decomposition and Principal Component Analysis:

Suppose that the empirical correlation matrix C has N eigenvalues $\lambda_1 \geq \dots \geq \lambda_k > \lambda_{\max}^{RMT} > \dots \geq \lambda_N$ with the corresponding eigenvectors $u_1, u_2, \dots, u_N$. The matrix C can be diagonalised as

$$C = U \Lambda U^{Tr} = \sum_{i=1}^N \lambda_i u_i u_i^{Tr}, \quad (9)$$

where $\Lambda = \text{diag}\{\lambda_1, \dots, \lambda_N\}$ is a diagonal matrix, and U (size $N \times N$) is the orthonormal matrix with i th column being the eigenvector u_i .

Defining $y_i = u_i^{Tr} \mathcal{X}$ as the i th principal component, we obtain

$$\text{Var}(u_i^{Tr} \mathcal{X}) = u_i^{Tr} \text{Cov}(\mathcal{X}) u_i = u_i^{Tr} C u_i = \lambda_i. \quad (10)$$

Hence,

$$\sum_{i=1}^N \text{Var}(y_i) = \sum_{i=1}^N \lambda_i = N = \sum_{i=1}^N \text{Var}(\mathcal{X}_i). \quad (11)$$

From equations (10) and (11), we can immediately see that $\frac{\lambda_i}{N}$ can be used as an indicator for the fraction of the total variance of the banking system in each market layer explained by the i th principal component $y_i = u_i^{Tr} \mathcal{X}$ (e.g. Jolliffe 1986, Kritzman *et al.* 2011, Wang *et al.* 2011, Billio *et al.* 2012, Gewers *et al.* 2018, Lux *et al.* 2020, Guerini *et al.* 2023, Luu 2021). Notice that since the eigenvalues are sorted in descending order, the first principal component $y_1 = u_1^{Tr} \mathcal{X}$ explains the greatest fraction of the total variance, the second principal component $y_2 = u_2^{Tr} \mathcal{X}$ in turn explains the greatest fraction of the leftover variance, and so forth (Kritzman *et al.* 2011, Billio *et al.* 2012).

‡ Note that when $\mathcal{X}$ is a heavy-tailed random matrix, its largest eigenvalues might exceed the upper bound $\lambda_{\max}^{RMT}$ implied by the Marchenko-Pastur law (e.g. see Biroli *et al.* 2007, Bouchaud and Potters 2015, Luu 2021, among others). However, our examination of this possibility reveals that the main conclusions remain largely unchanged. For the sake of conciseness, we, therefore, only report the results for the case with $\lambda_{\max}^{RMT}$ defined in equation (7).

Therefore, we can assume that the whole matrix C consists of different parts of correlations that are driven by various uncorrelated components representing different risk components. In particular, the first part $C^{sys.}$ is driven by the systematic component (i.e. namely the market mode) with a wider impact in each market layer (e.g. in the stock market or the credit market). The influence of such a common component on the correlations between banks can be inferred from the largest eigenvalue λ_1 and its corresponding eigenvector u_1 (Plerou *et al.* 2000, 2002):

$$C^{sys.} = \lambda_1 u_1 u_1^{Tr}. \quad (12)$$

Important signals or imprints of particular economic mechanisms can still be extracted from the consecutive eigenvalues of λ_1 , if they also deviate from the random bulk. If this is the case, we can further decompose C into three distinct parts $C^{sys.}$, $C^{gr.}$, and $C^{rand.}$ as

$$C = (C^{sys.} + C^{gr.}) + C^{rand.}, \quad (13)$$

where

$$C^{gr.} = \sum_{i=2}^k \lambda_i u_i u_i^{Tr}. \quad (14)$$

represents the effects of non-systematic components on some subsets of banks, and the rest

$$C^{rand.} = \sum_{i=k+1}^N \lambda_i u_i u_i^{Tr}. \quad (15)$$

is assumed to be the segment attributed to noise or negligible correlations. In equations (14) and (15), k is the number of the top largest eigenvalues that provide genuine information of the correlation structure, compared to the one explained by RMT.[†] It is also the number of the significant principal components explaining the greatest fractions of the total variance.

We shall now explain how to use the principal component analysis to infer the systematic risk and to examine the commonality among banks in each market layer. To this end, following Kritzman *et al.* (2011) and Billio *et al.* (2012), we define the absorption ratio AR_i ($i = 1, 2, \dots, N$) as

$$AR_i = \frac{\sum_{j=1}^i \text{Var}(y_j)}{\sum_{j=1}^N \text{Var}(\mathcal{X}_j)} = \frac{\sum_{j=1}^i \lambda_j}{N}. \quad (16)$$

Here, the value of each AR_i can be thought of as the portion of the total variance of $\{\mathcal{X}_j\}_{j=1}^N$ cumulatively explained by the first i principal components. Hence, from a perspective of risk management, $AR_1, \dots, AR_k$ illustrate the cumulative risk associated with the k significant principal components,

[†] Obviously, the smallest eigenvalues that are below the lower bound may also carry patterns of correlations. However, in our present work, these eigenvalues are often relatively small and the contribution of the associated principal components to the total variance is also relatively lower. Hence, we decide to focus on the extraction of genuine information and its potential economic footprints from the top largest eigenvalues.

and $AR_N = 1$ stands for 100% of the total risk of all banks contributed by all N components. In practice, we can compare these ratios to a certain critical threshold AR^* and/or examine their evolution over time to identify the periods of increased commonality and systemic risk (e.g. Pukthuanthong and Roll 2009, Kritzman *et al.* 2011, Billio *et al.* 2012, Zheng *et al.* 2012, Meng *et al.* 2014, Lux *et al.* 2020). For instance, a higher level of the first few absorption ratios $AR_1, AR_2, \dots$ indicates that banks tend to be more united under the effects of the first few risk components.

We are also interested in the distribution of the effects of each component on the different banks, and it can be shown that each eigenvector element indeed represents the factor loading for a particular bank (e.g. see Kritzman *et al.* 2011, Lux *et al.* 2020, Guerini *et al.* 2023). To examine the overall level of homogeneity in the different factor loadings for each component i th, we compute the associated Inverse Participation Ratio (IPR), which is given by

$$IPR_i = \sum_{j=1}^N u_{ij}^4, \quad (17)$$

where u_{ij} with $j = 1, 2, \dots, N$ are the N elements of the eigenvector u_i (e.g. Zyczkowski *et al.* 1992, Mirlin and Fyodorov 1993, Guhr *et al.* 1998, Plerou *et al.* 1999, 2002). Since under the normalised eigenvectors $\sum_{j=1}^N u_{ij}^2 = 1$ ($\forall i = 1, 2, \dots, N$), we have

$$1 \geq IPR_i \geq \frac{1}{N}, \quad \forall i = 1, 2, \dots, N. \quad (18)$$

In general, a larger value of IPR_i implies that fewer banks actually dominate in the i th component. Notice that $IPR_i = \frac{1}{N}$ if and only if all N factor loadings (in absolute terms) are completely identical (i.e. $u_{ij} = \sqrt{\frac{1}{N}}$ for all $j = 1, 2, \dots, N$). In contrast, under the most hierarchical structure, $IPR_i = 1$ if and only if only one bank contributes the eigenvector u_{ij} while the loadings of other banks are zero. Hence, for each eigenvalue, the associated IPR can reveal the number of components that contribute significantly to the corresponding eigenvector.

Network Analysis of Correlation Matrices:

In our study, we adopt the so-called Planar Maximally Filtered Graph (PMFG) algorithm (e.g. see Tumminello *et al.* 2005, Di Matteo *et al.* 2010, Aste *et al.* 2010) to extract the network from the correlation matrix in each market layer. More specifically, from the decomposition shown in equation (13), for each year, in every market layer, we construct the Planar Maximally Filtered Graph (PMFG) of $(C^{sys.} + C^{gr.}) = (C - C^{rand.})$ driven by all significant risk factors. In the next step, we analyse various important topological structures of the two inferred networks corresponding to the two layers and examine how they differently evolve over time. The main reason for using PMFG over methods like the Minimum Spanning Tree (MST) (Mantegna 1999) in our present work is that PMFG can capture the hierarchical organisation captured by MST while being able to retain more information by allowing 3- and 4-node cycles and closed motifs in the filtered network. This is crucial

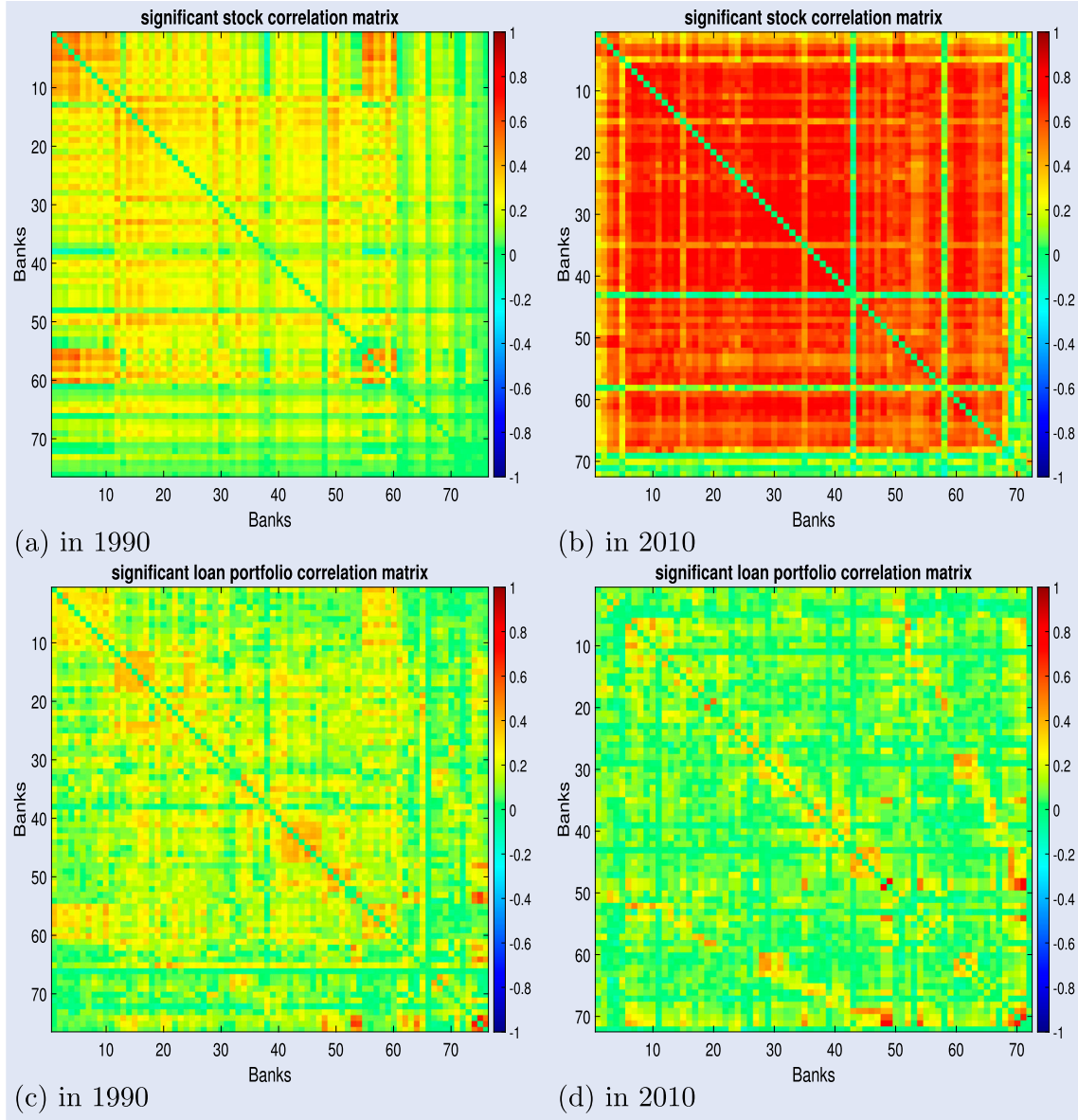


Figure 1. Different layers of significant correlations among Japanese banks, in 1990 and 2010 as examples. The colorbar indicates the correlation levels ranging from -1 to 1 . Note that in this figure, each year τ , the banks are arranged according to their indices from 1 to $N_B(\tau)$, rather than by their unique IDs or names, as mergers may cause banks to appear unevenly across different years. (a) in 1990. (b) in 2010. (c) in 1990 and (d) in 2010.

for assessing potential repercussive propagation among banks from a topological perspective (e.g. Bardoscia *et al.* 2016, Luu *et al.* 2021).

3. Multilayer architecture of correlations in the banking system of Japan

Here, we analyse the two layers of the correlations in the banking system of Japan, i.e. the loan correlations originating from the credit market and the stock return correlations arising from the equity market. Figure 1 depicts the matrices of significant correlations of the loan portfolio and stock returns of Japanese banks, in the two years 1990 and 2010 as

two examples.[†] Notice that these are the matrices in which the effects of the eigenvalues located inside the random bulk $[\lambda_{\min}^{RMT}, \lambda_{\max}^{RMT}]$ are filtered out. Overall, we find that the banking system exhibits different time-varying correlation structures in these two layers.[‡] This implies that the pairs of banks whose stock returns are highly aligned do not necessarily synchronise their loan provisions to non-financial firms in the credit market and vice versa. In other words, correlation risks

[†] For the sake of illustration, from figures 1 to 6, we focus on these two selected years. Results for the other years are available upon request.

[‡] The Spearman's correlation coefficients between the rank values of $\{C_{ij}^{Stock}\}$ and $\{C_{ij}^{loan}\}$ are 0.438^* and 0.1250^* in 2010, respectively. The results for the other years together with the Pearson correlation coefficients are reported in figure A4 in the Appendix.

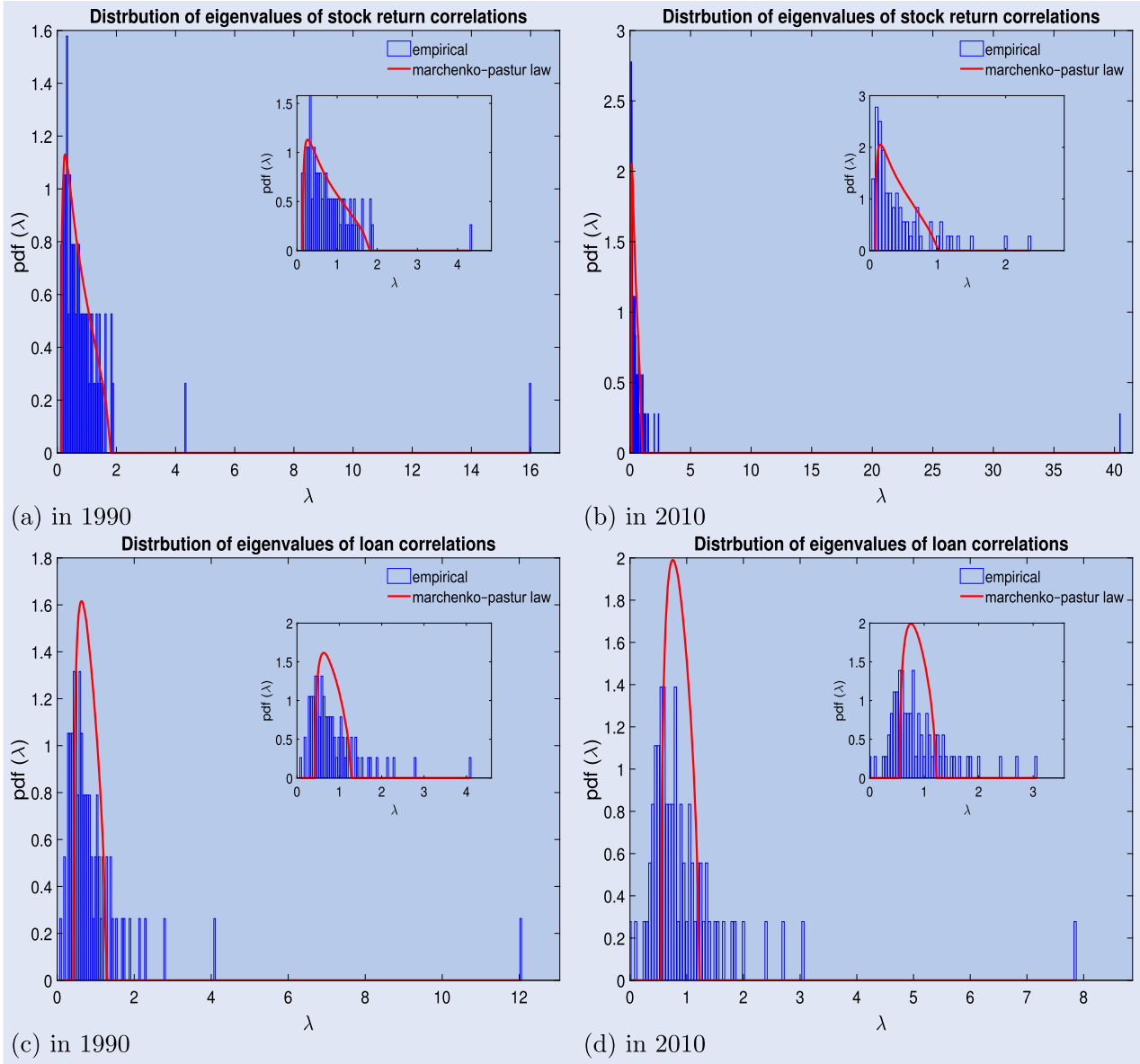


Figure 2. Distributions of the eigenvalues of the correlation matrices for loan portfolios and stock returns in 1990 and 2010, compared with the Marchenko-Pastur distribution. In each panel, the inset figure displays the distribution when the market mode (i.e. λ_1) is excluded. (a) in 1990. (b) in 2010. (c) in 1990 and (d) in 2010.

from the stock market may not fully capture or reflect those arising from the loan market. We can further decompose the contribution of the significant components to the correlation matrix by examining different parts associated with different types of the significant eigenvalues, e.g. the part driven by the eigenvalues exceeding $\lambda_{\max}^{RMT}$ and the rest contributed by the ones that are below $\lambda_{\min}^{RMT}$ as in figures A1 and A2 in the Appendix, respectively. As demonstrated, although some of the smallest eigenvalues deviate from the random bulk on the left hand side, their contributions to pairwise correlations appear to be negligible. In contrast, the eigenvalues beyond $\lambda_{\max}^{RMT}$ play a crucial role in shaping the main structure of correlations.

To identify the statistically significant components governing the correlations, we compare the eigenvalues of the empirical correlation matrices with those of the random

counterpart explained by the MP law shown in equation (7) (i.e. the theoretical distribution of eigenvalues of $C = \frac{1}{K} \mathcal{X} \mathcal{X}^T$, when the elements $X_{ij} \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$). In all years, we observe that only few eigenvalues are beyond the upper bound predicted by the MP law,[†] while the majority of eigenvalues are in the random bulk or less than the lower bound $\lambda_{\min}^{RMT} = (1 - \sqrt{1/Q})^2$. This shows that only a few principal components play a vital role in explaining the correlation dynamics and contain genuine information about the structure of interdependencies among banks. Nevertheless, it is worth mentioning that there exist more significant eigenvalues (i.e. those beyond the upper bound) in loan correlations

[†] This result still holds when we take into account the possible effects of stricter bounds under the presence of extreme values of X_{ij} (e.g. see Biroli et al. 2007, Lux et al. 2020).

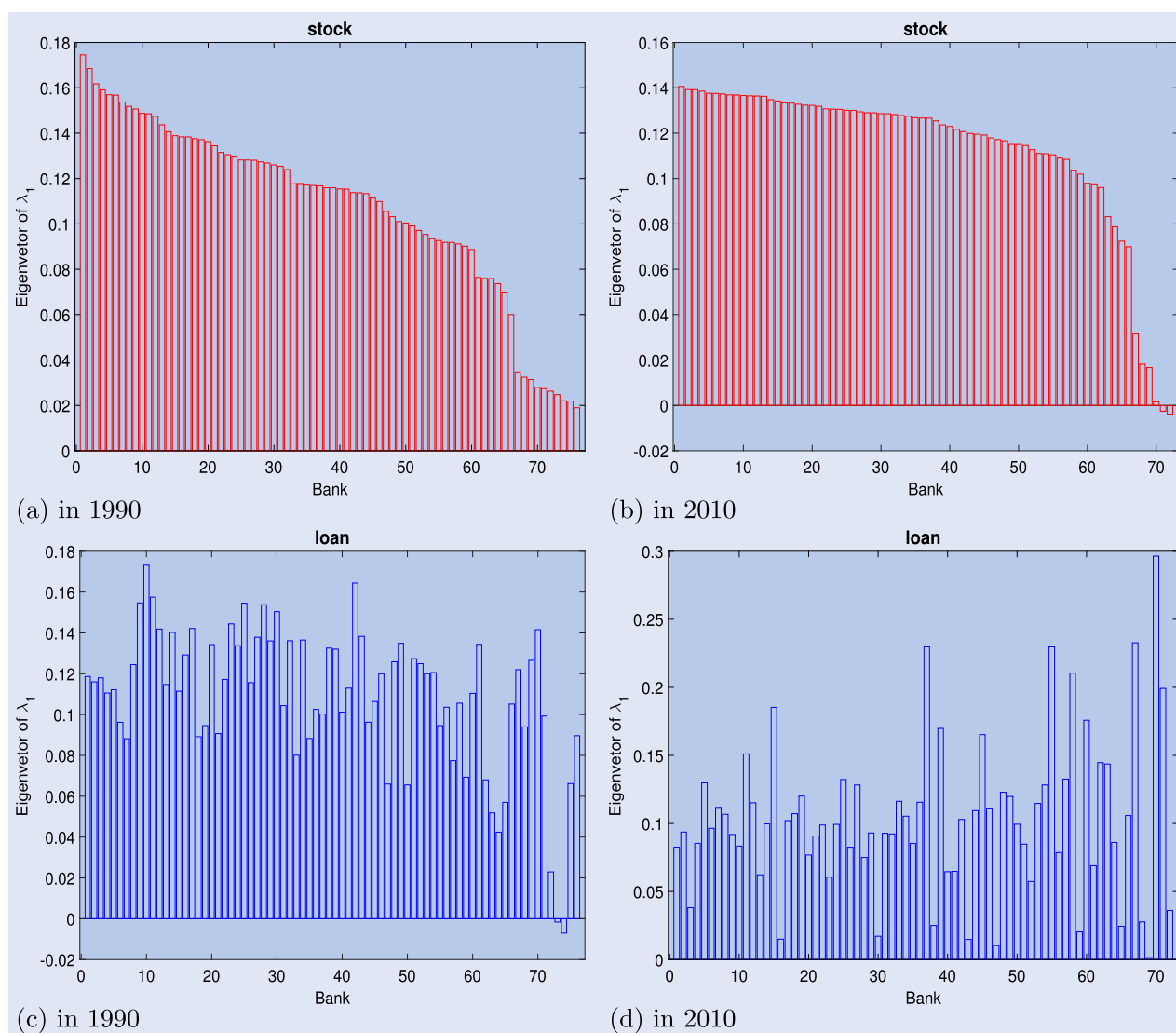


Figure 3. Eigenvector elements of λ_1 in 1990 and 2010. Banks are in descending order of the eigenvector elements of λ_1 in the stock return correlation layer in each year. The Spearman correlation coefficients of the orders are 0.423 and -0.095, respectively for the two example years 1990 and 2010. (a) in 1990. (b) in 2010. (c) in 1990 and (d) in 2010.

than in stock return correlations[†] (see also figure A5 in the Appendix), suggesting that the two markets exhibit different degrees of homogeneity and that, in particular, there are more significant factors driving the correlations among banks in the credit market than in the stock market (figure 2).

It should be noted that, besides the top eigenvalues, there are also small ones deviating from the random bulk, i.e. those that are smaller than $\lambda_{\min}^{RMT}$. These eigenvalues, of course, may also carry genuine information on the interdependencies among banks. We further compute the contributions of these small and significant eigenvalues to the total variance. In general, we find that their aggregated and individual contributions are relatively small for the stock market. In contrast, for the credit market, while these eigenvalues may have negligible individual roles, their aggregated

contribution to the total variance can be more significant. This again confirms that there can be a more diverse set of components underlying the correlations in the latter market layer.

After a first glance on the distinct features between the two market layers shown in figure 1, we shall now conduct a more in-depth examination of the order of eigenvector elements corresponding to the top largest eigenvalues. Notice that from a PCA perspective, each of the elements of the i th eigenvector, all else being equal, indicates the impact of the associated i th component on a particular bank. Figure 3 demonstrates the comparison between the order of eigenvector elements associated with the largest eigenvalue of C^{Stock} and that of eigenvector elements associated with the largest eigenvalue of C^{Loan} in the two example years. Our results typically show that the effects have a dissimilar rank in the different layers. Such a difference is still observed in the eigenvector elements (in absolute terms) associated with the subsequent largest

[†] For most of the years, the number of significant largest eigenvalues in loan correlations is more than two times larger than that in stock return correlations.

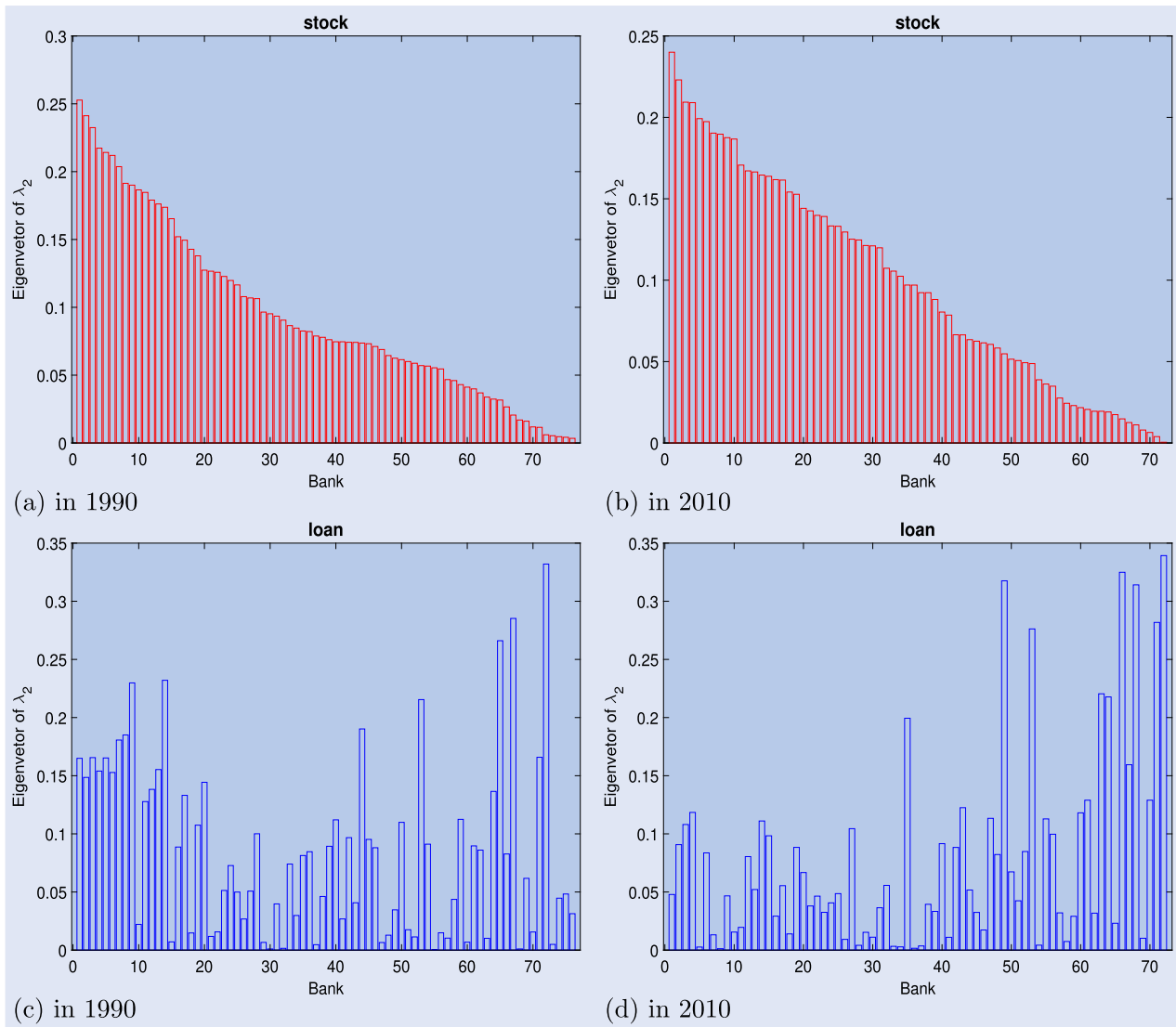


Figure 4. Eigenvector elements (in absolute terms) of λ_2 , in 1990 and 2010. Banks are in descending order of the eigenvector elements of λ_2 in the stock return correlation layer. The Spearman correlation coefficients of the orders are 0.266 and -0.314 , respectively for the two example years 1990 and 2010. (a) in 1990. (b) in 2010. (c) in 1990 and (d) in 2010.

eigenvalues (see figures 4 and 5).[†] Further comparison of the eigenvector elements for other pairs, such as those associated with λ_2 in the stock market and λ_1 in the credit market, unveils distinct rank structures. Hence, the non-systematic component within one layer also appears to exhibit weak alignment with the market mode in another layer.[‡] This again confirms the distinct structures of correlation risks originating from the two market layers.

In the next step, we further examine whether the components inferred from the correlation matrix in each market layer

[†] Notice that if u_i is an eigenvector corresponding to λ_i , then $-u_i$ is also an eigenvector. While almost all elements of u_1 typically share the same sign, this is generally not the case for those of $u_2, u_3, \dots$. Therefore, we take the absolute values of the elements in the subsequent eigenvectors and compare their rankings.

[‡] The Spearman correlation coefficients of the orders for $\{u_{2j}\}_{j=1}^N$ in the former and $\{u_{1j}\}_{j=1}^N$ in the later (in absolute terms) are -0.010 in 1990 and -0.134 in 2010, respectively.

have a pervasive influence on many banks or only affect a few banks. To this end, we measure the Inverse Participation Ratio (IPR) based on the eigenvector elements of each eigenvalue (see equation (17)) and compare it with the overall average of all IPRs. Recall the range for IPRs in equation (18), i.e. $\text{IPR} = \frac{1}{N}$ if and only one if all the factor loadings are completely homogeneous, while $\text{IPR} = 1$ if only one component completely dominates the others. Furthermore, eigenvectors in the random bulk $[\lambda_{\min}^{\text{RMT}}, \lambda_{\max}^{\text{RMT}}]$ typically have Gaussian distributed components while the deviations from the RMT prediction of a Gaussian tend to be more pronounced as the separation from the RMT upper bound $(\lambda_k - \lambda_{\max})$ increases. Therefore, following Plerou *et al.* (1999, 2002), in figure 6 we examine the relationship $\text{IPR}(\lambda)$ vs. λ and compare the IPR values with their average instead of with $\frac{1}{N}$. Similar to the results for the two example years shown in figure 6, we typically find that the IPR ratios corresponding to the two largest eigenvalues λ_1 and λ_2 are relatively small, implying that the first and the second

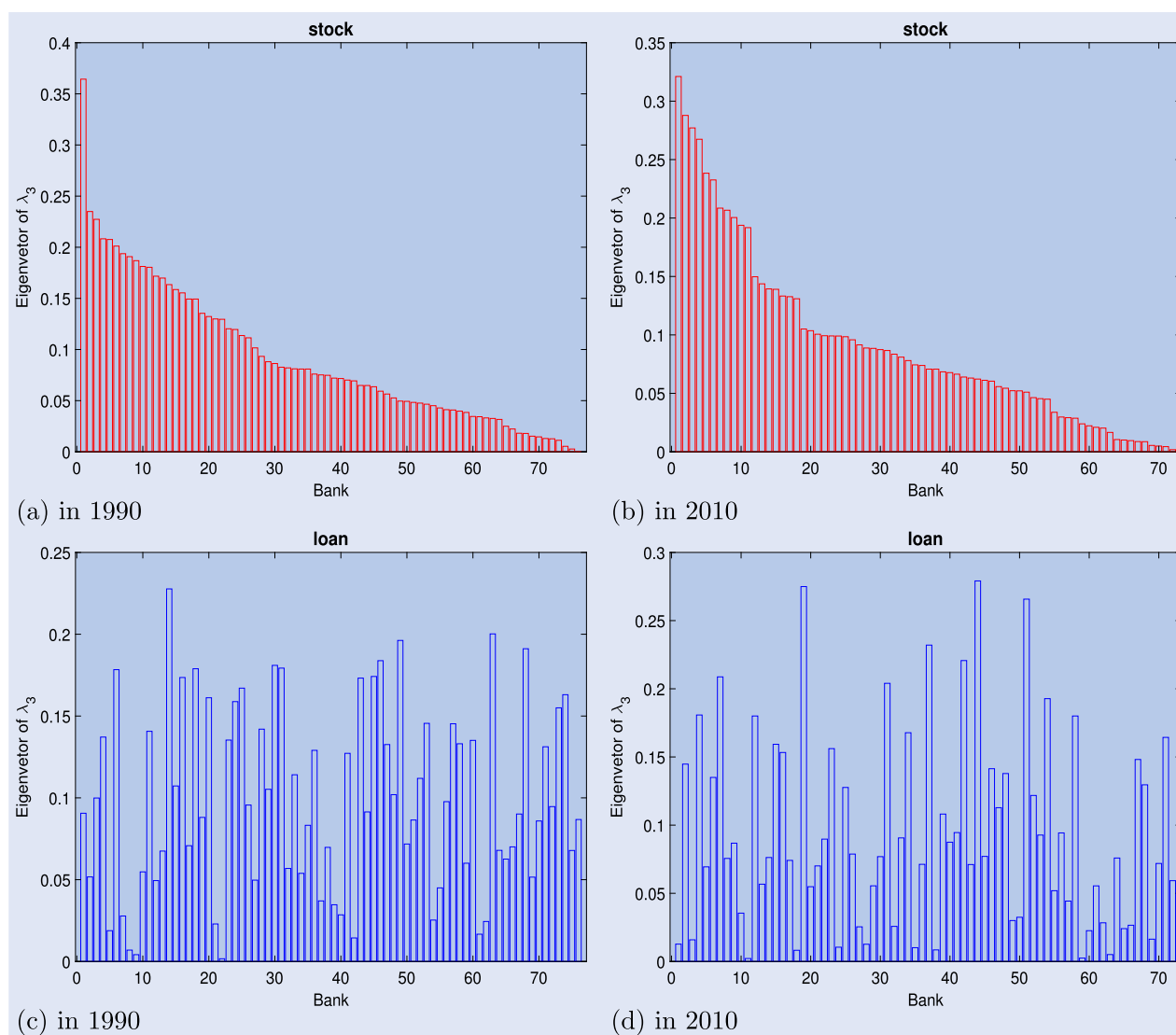


Figure 5. Eigenvector elements (in absolute terms) of λ_3 , in 1990 and 2010. Banks are in descending order of the eigenvector elements of λ_3 in the stock return correlation layer. The Spearman correlation coefficients of the orders are -0.070 and 0.063 , respectively for the two example years 1990 and 2010. (a) in 1990. (b) in 2010. (c) in 1990 and (d) in 2010.

strongest factors do have a widespread effect in both markets. Meanwhile, some other IPRs of the next eigenvectors may be slightly or significantly above the average. This signals a certain level of localised behaviour in these eigenvectors, meaning that the associated factors have a stronger impact on some particular subgroups of banks.

We shall now explore the systemic risk and the overall commonality among banks in the two layers using the methods of principal component analysis and absorption ratios. The time-series graph of the top five eigenvalues, the corresponding absorption ratios and the absorption ratio of all significant principal components are shown in figure 7 for loan correlations and in figure 8 for stock correlations.

In the credit market, via the inspection of the different largest eigenvalues and the contribution of the corresponding components to the total variance of loans, we typically observe distinct dynamics in different periods: from 1982 (the beginning of our sample) to 1990 (when the bubble

started to burst), the lost decade from 1991 to 2000, the period from 2001 prior to the global financial crisis and the aftermath. As shown in figure 7, the largest eigenvalue experiences an increase during the time of the economic bubble in Japan, reaching its peak in 1990. Subsequently, it witnesses a decrease during the lost decade 1991–2000, corresponding to the period of the economic bubble burst. It then starts to increase again from 2001 to 2007 and significantly fluctuates in 2008 and 2009. The next four eigenvalues also exhibit a similar trend (although their changes over the years are blurred compared with the changes in the largest eigenvalue in figure 7). In contrast, as illustrated in the corresponding absorption ratios from AR_1 to AR_5 , overall, the relative contribution of the factors associated with these top five largest eigenvalues declines from the beginning of the sample to 2000, and slightly increases from 2001 to the global financial crisis, before similarly showing a large fluctuation in 2008 and 2009. The decline of the contribution by the top five

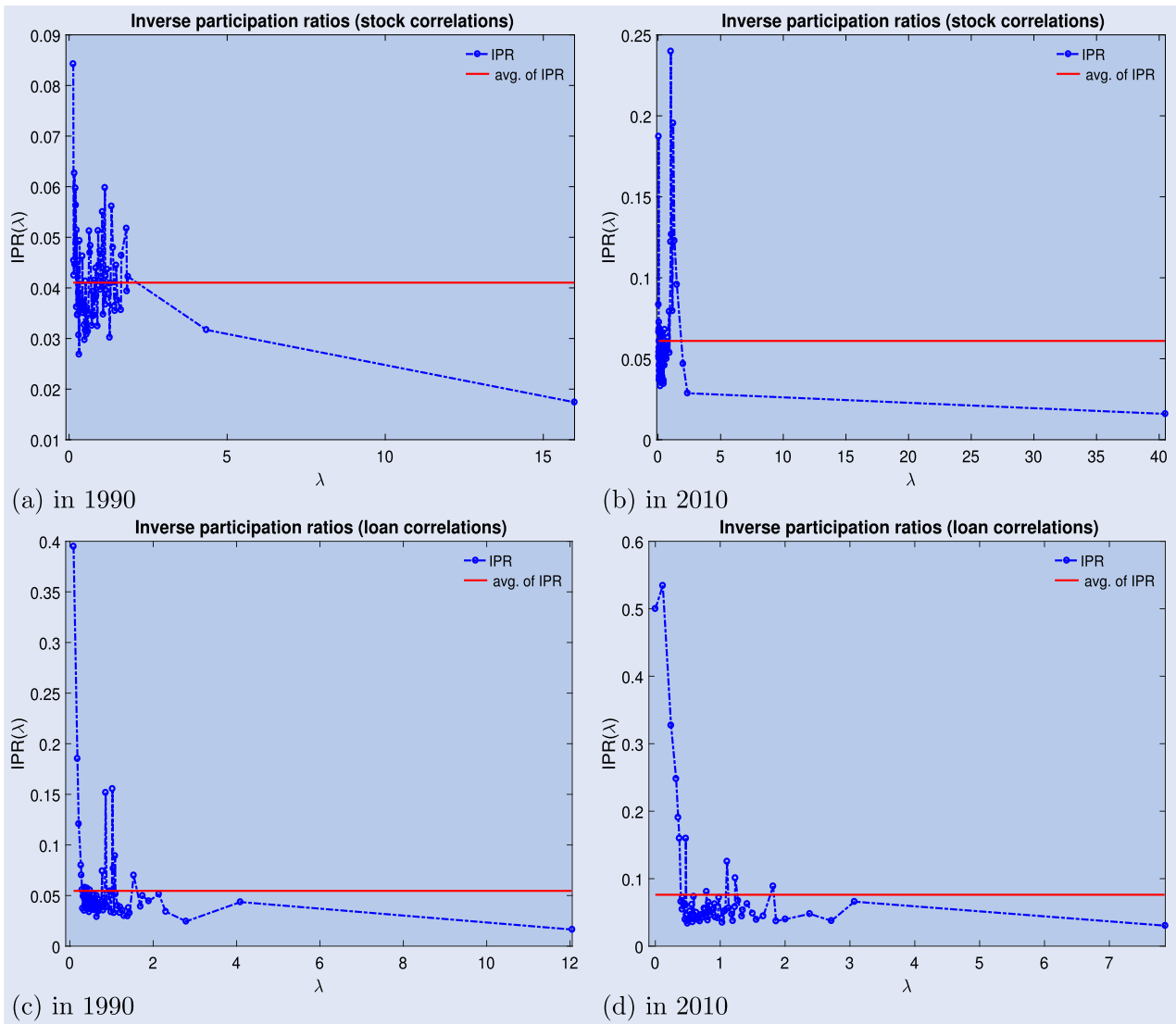


Figure 6. Banks' contribution to components based on Inverse Participation Ratio, in 1990 and 2010. In these four panels, we plot the IPR vs. the corresponding eigenvalue λ . (a) in 1990. (b) in 2010. (c) in 1990 and (d) in 2010.

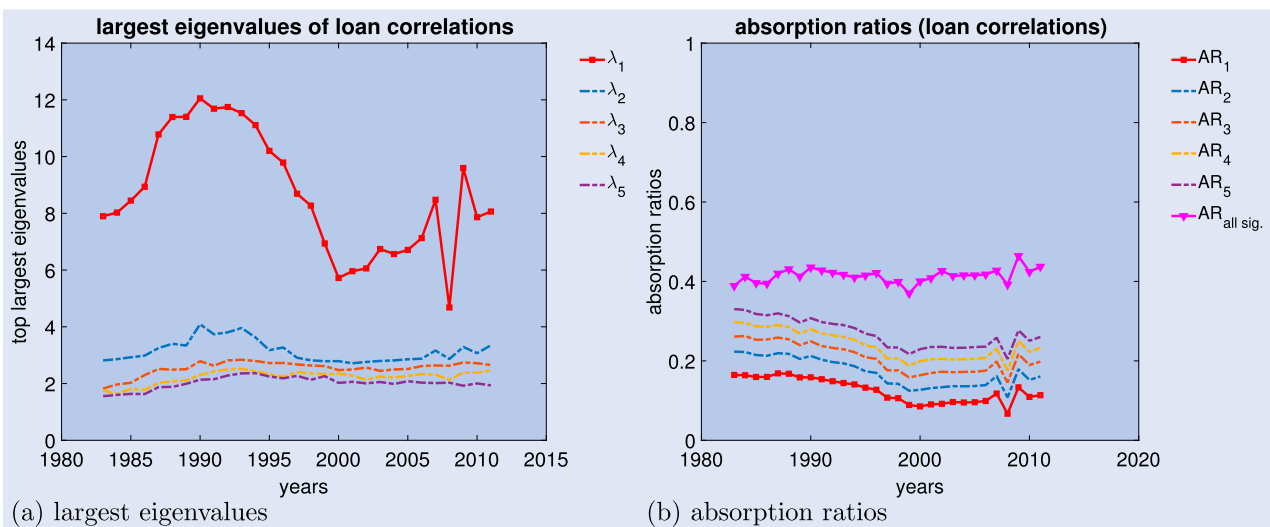


Figure 7. Evolution of the largest eigenvalues and the contribution of the corresponding components to the total variance in the credit market over the period from 1982 to 2011. (a) largest eigenvalues and (b) absorption ratios.

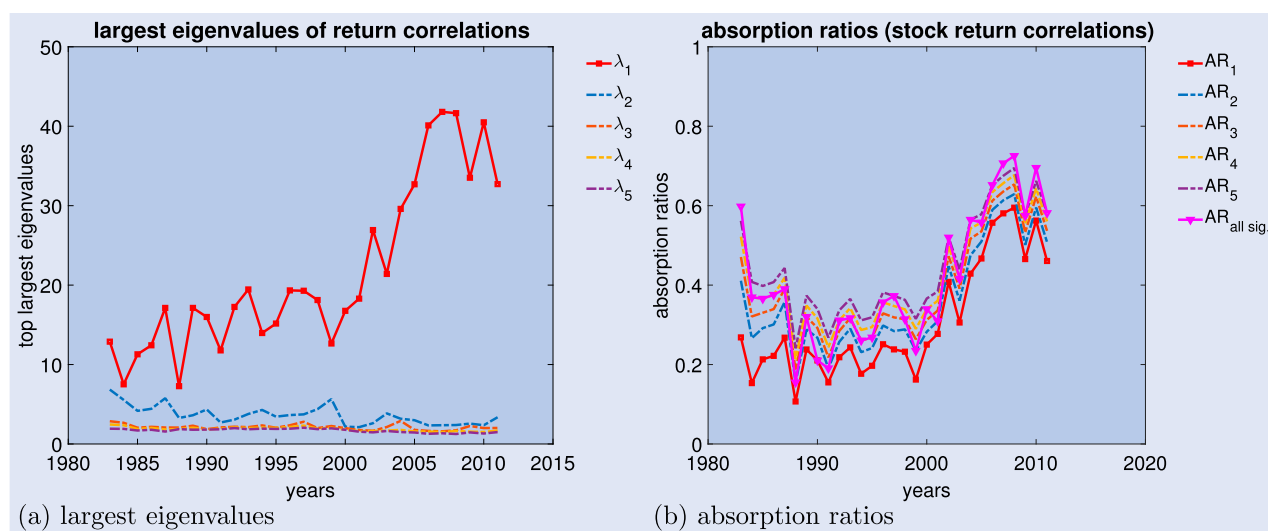


Figure 8. Evolution of the largest eigenvalues and the contribution of the corresponding components to the total variance in the equity market over the period from 1982 to 2011. (a) largest eigenvalues and (b) absorption ratios.

eigenvalues in the period prior to 1991 is due to the fact that although in absolute terms each of the risk factors associated with these eigenvalues becomes stronger during the bubble time the other factors (including the noise ones) also have a larger impact. Nevertheless, $AR_{all\,sig.}$, i.e. the *aggregated influence of all significant factors*, is somewhat more stable, staying around 40–50 (%) of loan variation in the credit market during the whole sample. Of this fraction, the strongest common factor (i.e. the market mode) alone explains only 10–20 (%).

The analysis of correlations originating from the equity market, on the other hand, documents different results of systemic risk and overall commonality among banks. In comparison to the credit market layer, prior to the year 2000, the top five largest eigenvalues vary significantly, causing similar dynamics in the absorption ratios from AR_1 to AR_5 . After that, they increase subsequently over most of the years before 2008. Notably, this change is led by the largest eigenvalue. The contribution of the market mode (i.e. the global strongest factor) alone to the total variance in stock returns become more and more important, peaking at about 60% in 2007, while the aggregated effect of all significant factors is just slightly higher (about 70%). This implies that the equity market becomes more unified under the influence of the global strongest factor, signalling a higher level of systemic risk.

In the final step, we apply the PMFG algorithm to infer the network connectivity from the two correlation matrices. Our aims are to further identify the topological skeleton of the (statistically) significant segment of the correlation network in each layer and to examine how the structure of this network has evolved over the years. We show the visualisation of the PMFG of $(C^{sys.} + C^{gr.}) = (C - C^{rand.})$ in the two selected years 1990 and 2010 (see figure 9, and also figures A6 to A9 in the Appendix for the other selected years). In these figures, we report the names of the five banks (hubs) that have the largest degrees in the inferred network. The results confirm what we have observed from the principal component and eigenvector

analyses: Banks acting as the hubs in one correlation network layer are not necessarily the key players in the other one and vice versa. On top of that, although there is a certain degree of persistence over several consecutive years, the hubs also change in different periods.

Furthermore, computing the important network statistics, we find that the two market layers exhibit different levels of clusterings, assortativities and shortest distance averages among banks over the years (see figures A10 to A13 in the Appendix).

4. Dynamics of correlations and banks' characteristics

4.1. Estimation methodology

The results summarised in Section 3 have revealed different correlation structures and temporal dynamics in the credit and equity markets. We now further explore such differences by analyzing how the correlations stemming from the two markets reflect the fundamental information and characteristics of banks. To this end, first, we measure the qualitative and quantitative similarities between banks. Next, we examine whether they are related to future co-movements in credit provisions or in stock returns.

Denote the correlation matrix in year τ by $C(\tau) = \{C_{ij}(\tau)\}_{N_B \times N_B}$ ($1983 \leq \tau \leq 2011$ in the dataset we considered).[†] We consider a statistical model such that each pairwise correlation $C_{ij}(\tau)$ at time τ is a linear function of the previous correlation $C_{ij}(\tau - 1)$, and of the qualitative and quantitative similarities between bank i and bank j under various criteria. More specifically, we assume in the hypothesised models that the qualitative similarities include the similarities in

[†] Notice that $C(\tau)$ can be the stock correlation matrix $C_{Stock}(\tau)$ or the loan correlation matrix $C_{Loan}(\tau)$. We will separately run the regressions mentioned in equations (19) and (20) for the two different market layers.

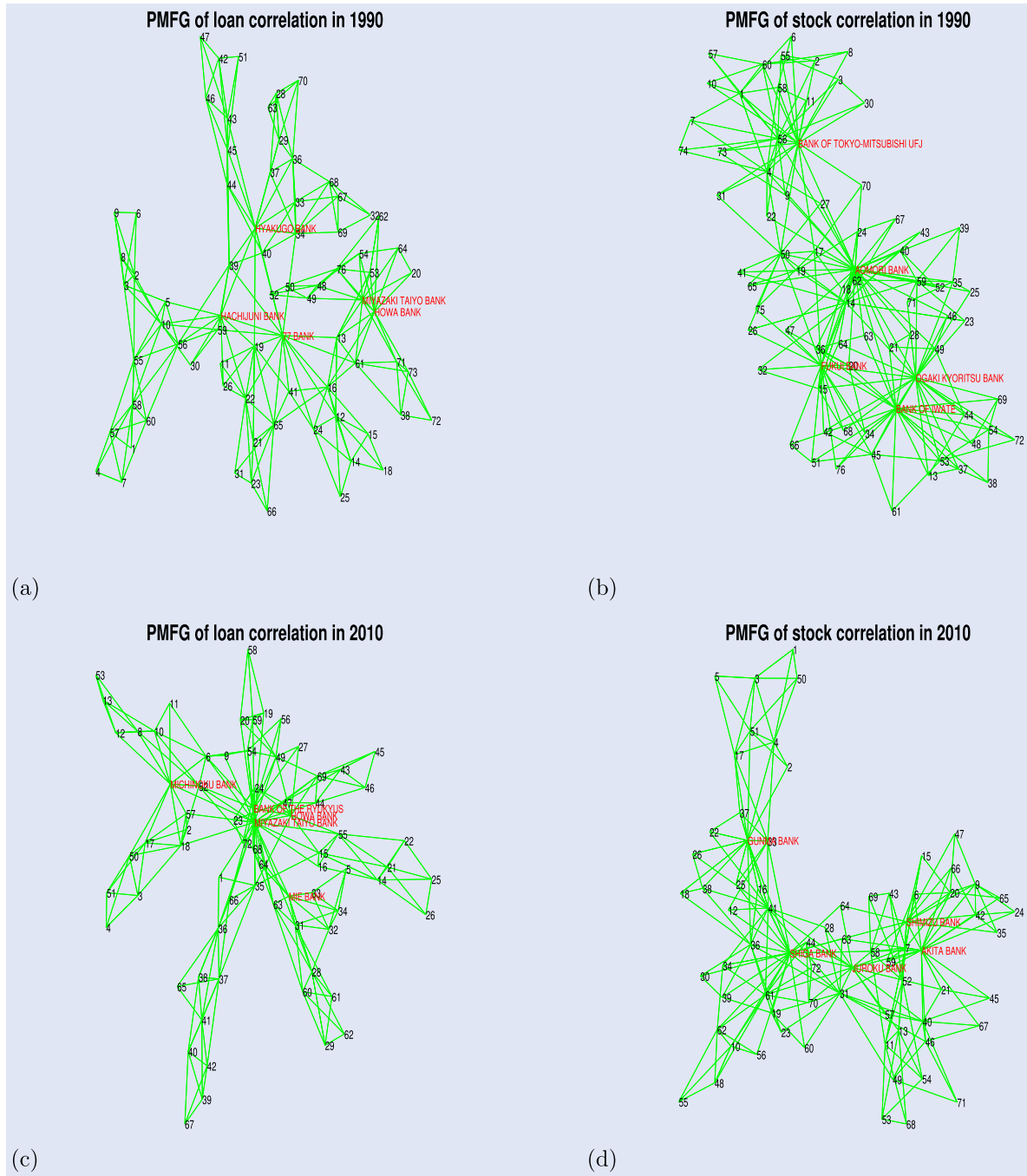


Figure 9. PMFG networks of the correlations in 1990 and 2010.

bank types and bank locations. The quantitative counterparts indicate the degree of similarity in sizes, equities, deposits, amounts of cash, securities, total loans, and incomes. These different criteria of characteristics and fundamental information have been often documented as potentially important determinants of co-movements of stock returns in the pertinent literature.[†]

Let us assume in general that at time $(\tau - 1)$ we have n different similarity criteria, and $Similarity_{ij}^{(criterion_k)}(\tau - 1)$ represents the degree of similarity between bank i and bank j based

on the k th criterion, for $k = 1, 2, \dots, n$ (see the Appendix for further explanations). We start with the following hypothesised model f_1 for the raw correlations

Model 1:

$$C_{ij}(\tau) = f_1(C_{ij}(\tau - 1), Similarity_{ij}^{(criterion_1)}(\tau - 1), \dots, Similarity_{ij}^{(criterion_n)}(\tau - 1)) + \epsilon_{ij}(\tau) \quad (19)$$

which we assume to be linear in the following investigation. However, as shown in equations (12), (13) and (14), the raw correlation matrix can be decomposed into three parts driven by the effects of the *systematic common factor (or market mode)*, the *non-systematic factors* and *noise*. If we

[†] See, for example, Shiller (1989), Pindyck and Rotemberg (1990), Barberis and Shleifer (2003), Barberis et al. (2005), Box (2018) and Do et al. (2022) and the literature therein for the discussion of various potential determinants of the financial correlations.

Table 1. Model 1: Yearly similarities and stock return raw correlations from 1983 to 1990.

Variables/Years	1983	1984	1985	1986	1987	1988	1989	1990
$z_{ij,Stock}(\tau - 1)$	0.0069 (0.2626)	0.1922* (15.2208)	0.2308* (11.8364)	0.2502* (17.1315)	0.5367* (26.7751)	0.1828* (21.5095)	0.3773* (18.8183)	0.4330* (35.2602)
size sim.	0.0430 (1.8749)	0.1844* (14.4456)	0.1422* (10.9979)	0.0578* (5.7173)	0.0787* (6.7743)	0.0355* (5.1176)	0.0392* (4.3252)	0.0800* (10.2248)
cash sim.	0.1128* (3.3800)	0.0251 (1.2385)	0.0196 (1.0419)	0.0161 (1.1907)	0.0918* (5.4963)	-0.0031 (-0.3152)	0.01877 (1.7905)	0.0330* (4.1684)
security sim.	-0.0342 (-0.8522)	0.0195 (0.8664)	-0.0245 (-1.0977)	0.1236* (6.4149)	-0.1949* (-8.3217)	-0.0134 (-0.9862)	0.0673* (3.7022)	-0.0723* (-4.8770)
total loan sim.	0.0123 (0.2080)	0.3190* (7.3839)	0.0204 (0.5086)	0.2196* (6.1983)	-0.0059 (-0.1250)	0.0431 (1.4576)	-0.0109 (-0.3350)	-0.0280 (-1.1701)
deposit sim.	-0.1075** (-2.2375)	-0.0413 (-1.6659)	-0.1569* (-6.0236)	0.1217* (5.8724)	-0.2068* (-8.3220)	0.0017 (0.1041)	0.01166 (0.5509)	0.0358 (1.9142)
equity sim.	0.0801 (1.3139)	-0.0217 (-0.6838)	-0.1169* (-3.7479)	0.1040* (4.6942)	-0.0318 (-1.2319)	0.0074 (0.4660)	0.0676* (3.2973)	-0.0284 (-1.5632)
income sim.	-0.1921* (-3.9537)	0.0017 (0.0735)	-0.1151* (-4.8866)	-0.0499* (-2.7276)	-0.0462** (-2.3528)	-0.0193 (-1.6794)	0.1317* (7.7051)	-0.0248 (-1.6256)
location sim.	0.1221* (7.3051)	0.1921* (19.0923)	0.0717* (6.6247)	0.1509* (18.2674)	0.1457* (13.6309)	0.1198* (19.0915)	0.0286* (3.4317)	0.0818* (12.6562)
type sim.	0.0798* (4.0374)	-0.0505* (-3.9894)	-0.0033 (-0.2598)	-0.0585* (-6.2736)	-0.0176 (-1.5386)	-0.0112 (-1.9549)	0.0030 (0.4147)	-0.0493* (-8.2703)
observations	1892	2070	2256	2756	3080	4032	4290	4830
adj. R-squared	0.1608	0.4655	0.2892	0.3708	0.4339	0.2962	0.1711	0.2973

Notes: The dependent variable $z_{ij,Stock}(\tau)$ is the Fisher transformation of the Pearson correlation coefficient $C_{ij,Stock}(\tau)$. The model $z_{ij,Stock}(\tau) = \beta_0 z_{ij,Stock}(\tau - 1) + \sum_{k=1}^9 \beta_k \text{Similarity}_{ij}^k(\tau - 1) + \epsilon_{ij}(\tau)$ is estimated by OLS for each year τ . We report the coefficients for the lagged dependent variable as well as for the nine balance-sheet-based similarity variables in the table below with t-statistics in parentheses. * and ** indicate statistical significance at the 1% and 5% levels, respectively.

Table 2. Model 1: yearly similarities and stock return raw correlations from 1991 to 2000.

	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
$z_{ij,Stock}(\tau - 1)$	0.4857* (46.5507)	0.9535* (63.6143)	0.6091* (70.8051)	0.5748* (66.4861)	0.5652* (54.7357)	0.6541* (58.545)	0.6057* (58.2021)	0.5353* (49.7222)	0.4663* (45.2471)	0.6534* (38.8044)
size sim.	0.0380* (6.7279)	0.0383* (5.3266)	0.0714* (11.9371)	0.0489* (8.8863)	0.0748* (12.7055)	0.0820* (13.4303)	0.0313* (5.5708)	0.0789* (12.5159)	0.0969* (16.8049)	0.1698* (21.2828)
cash sim.	-0.0201* (-3.3106)	-0.0386* (-4.7089)	-0.0197* (-2.9389)	0.0311* (5.0641)	-0.0252* (-4.0179)	-0.0018 (-0.2758)	0.0123 (1.8517)	-0.0165** (-2.5021)	0.0320* (4.963)	-0.0645* (-7.0457)
security sim.	-0.0299** (-2.5112)	0.0073 (0.4437)	-0.0519* (-4.2603)	-0.0204 (-1.6244)	0.0094 (0.7221)	-0.0023 (-0.1622)	0.0816* (5.8432)	0.0826* (6.5675)	-0.0116 (-0.9351)	0.2286* (11.7754)
total loan sim.	0.1792* (9.5545)	0.1096* (4.2968)	0.1176* (4.2871)	0.0840* (3.2448)	0.0418 (1.3838)	0.2217* (6.4791)	0.1912* (5.4007)	0.0293 (0.8657)	0.1765* (5.7085)	-0.4665* (-9.2184)
deposit sim.	-0.0245 (-1.6520)	-0.0771* (-4.2974)	-0.0718* (-5.4304)	0.0300* (2.6327)	0.0120 (1.0582)	-0.0392* (-3.2648)	-0.0398* (-3.5251)	0.0470* (3.2792)	-0.0145 (-1.1512)	0.1219* (2.7271)
equity sim.	-0.0596* (-3.6254)	-0.1256* (-5.7413)	0.1047* (5.9535)	0.0020 (0.1275)	0.0287 (1.7304)	-0.0720* (-4.6326)	0.0342** (2.4533)	0.0978* (8.2595)	0.0905* (9.4528)	-0.0279* (-2.7427)
income sim.	0.0639* (4.9610)	0.0002 (0.0247)	-0.0003 (-1.3988)	0.0567* (10.4893)	0.0424 (9.8812)	0.0000 (1.3338)	-0.0198* (-5.9901)	0.0002* (2.6930)	0.0001 (0.3495)	0.0040 (1.5021)
location sim.	0.1034* (18.5896)	0.0823* (10.8301)	0.0906* (14.0993)	0.1144* (19.0712)	0.0773* (12.1635)	0.1013* (14.4590)	0.0949* (13.3785)	0.1647* (22.6745)	0.1482* (21.5022)	-0.0338** (-2.5231)
type sim.	-0.0700* (-14.5258)	-0.0329* (-5.0852)	-0.0287* (-5.5559)	-0.0425* (-9.0128)	-0.0714* (-14.7330)	-0.0584* (-10.4297)	-0.0416* (-9.2076)	-0.0713* (-11.3660)	-0.0297* (-5.5729)	-0.0720* (-5.3280)
observations	5402	5550	6162	6162	5700	5550	5852	5852	5402	4422
adj. R-squared	0.4601	0.5564	0.5784	0.5763	0.5380	0.5641	0.5193	0.5486	0.5383	0.4565

Notes: The dependent variable $z_{ij,Stock}(\tau)$ is the Fisher transformation of the Pearson correlation coefficient $C_{ij,Stock}(\tau)$. The model $z_{ij,Stock}(\tau) = \beta_0 z_{ij,Stock}(\tau - 1) + \sum_{k=1}^9 \beta_k \text{Similarity}_{ij}^k(\tau - 1) + \epsilon_{ij}(\tau)$ is estimated by OLS for each year τ . We report the coefficients for the lagged dependent variable as well as for the nine balance-sheet-based similarity variables in the table below with t-statistics in parentheses. * and ** indicate statistical significance at the 1% and 5% levels, respectively.

exclude the effects of the systematic common factor, the residual correlations $C_{ij}^{fil,sys.}(\tau) = C_{ij}(\tau) - C_{ij}^{sys.}(\tau)$ are obtained. As before, we may now formulate a model for these filtered correlations. Since we only remove a systematic aggregate

factor, individual factors such as our bilateral similarity criteria may still play an important role in explaining these correlations. Hence, in the second model, we hypothesise that the residual correlations are again driven by their lag, the

Table 3. Model 1: Yearly similarities and stock return raw correlations from 2001 to 2011.

Variables/Years	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
$z_{ij,Stock}(\tau - 1)$	0.6032* (47.9565)	0.9415* (70.6183)	0.6239* (70.1587)	0.8695* (77.7455)	0.7833* (76.4666)	0.9326* (120.6364)	1.0099* (152.4397)	0.8300* (128.9408)	0.8584* (103.3838)	0.9041* (117.0457)	0.6072* (79.1705)
size sim.	0.1264* (15.8122)	0.0772* (9.3537)	0.1020* (14.9694)	0.0560* (7.3844)	-0.0361* (-4.8477)	0.0817* (13.4452)	0.0712* (12.6175)	0.0228* (3.9557)	0.0689* (10.8723)	0.0362* (5.1930)	0.1451* (19.7214)
cash sim.	-0.0309* (-3.6789)	-0.0179** (-2.0308)	-0.0045 (-0.6376)	-0.0257* (-3.4622)	0.0376* (4.4443)	0.0030 (0.4390)	-0.0185* (-2.6646)	0.0105 (1.4007)	-0.0143 (-1.8436)	-0.0220* (-2.6523)	0.0053 (0.6278)
security sim.	0.2039* (10.9153)	0.1351* (7.0987)	0.0849* (6.6203)	0.0022 (0.1625)	0.1058* (6.7350)	0.0998* (7.0272)	0.0266 (1.7925)	0.0276** (2.0514)	-0.0129 (-0.8400)	0.0704* (4.1643)	0.0353 (1.9173)
total loan sim.	-0.3358* (-6.7621)	-0.1820* (-3.2334)	-0.1500* (-3.9097)	0.0327 (0.8390)	-0.2745* (-6.7961)	-0.0232 (-0.6106)	-0.2484* (-6.9680)	0.0996* (3.0103)	-0.0710 (-1.7615)	-0.2102* (-5.4092)	-0.0891 (-1.9125)
deposit sim.	0.3728* (3.8697)	0.1800 (1.5435)	0.0064 (0.1434)	-0.1965* (-7.6747)	0.1541* (5.2071)	0.0597* (2.5866)	-0.1230* (-6.1137)	-0.0229 (-1.1698)	0.0322 (1.2832)	-0.1838* (-6.9292)	0.0502 (1.4814)
equity sim.	0.0366** (1.9973)	0.0215 (1.2882)	0.0236** (2.4549)	-0.0559* (-4.1587)	0.0680* (4.7260)	0.0943* (9.6500)	0.0650* (5.9777)	0.0183 (1.6564)	0.1122* (9.7056)	0.0113 (0.7720)	0.1614* (10.9426)
income sim.	0.0008** (1.9742)	-0.0006* (-2.7671)	-0.0001 (-0.3243)	0.0006* (4.0601)	0.0008* (4.4764)	0.0004** (2.5029)	0.0011* (5.6901)	-0.0000 (-0.2918)	-0.0000 (-1.6149)	0.0001 (1.1613)	0.0002* (3.8950)
location sim.	0.0286** (2.2568)	0.0038 (0.2995)	0.0476* (4.6797)	-0.0272** (-2.4728)	0.0161 (1.4378)	0.0031 (0.3144)	0.0239* (2.6434)	0.0329* (3.7517)	0.0217** (2.1457)	-0.0129 (-1.1673)	0.0322** (2.5238)
type sim.	-0.1364* (-3.8864)	-0.0325 (-0.8847)	-0.0095 (-0.9194)	-0.0113 (-1.5971)	0.0267* (3.2448)	-0.0640* (-9.6671)	-0.0121** (-2.0571)	-0.0287* (-4.9757)	-0.0004 (-0.0643)	0.0933* (14.3865)	-0.0753* (-9.8135)
observations	4032	3906	4290	4556	4422	4830	4970	4830	4692	5112	4830
adj. R-squared	0.5806	0.7147	0.7142	0.6838	0.6658	0.8237	0.8636	0.8061	0.7485	0.7858	0.6782

Notes: The dependent variable $z_{ij,Stock}(\tau)$ is the Fisher transformation of the Pearson correlation coefficient $C_{ij,Stock}(\tau)$. The model $z_{ij,Stock}(\tau) = \beta_0 z_{ij,Stock}(\tau - 1) + \sum_{k=1}^9 \beta_k \text{Similarity}_{ij}^k(\tau - 1) + \epsilon_{ij}(\tau)$ is estimated by OLS for each year τ . We report the coefficients for the lagged dependent variable as well as for the nine balance-sheet-based similarity variables in the table below with t-statistics in parentheses. * and ** indicate statistical significance at the 1% and 5% levels, respectively.

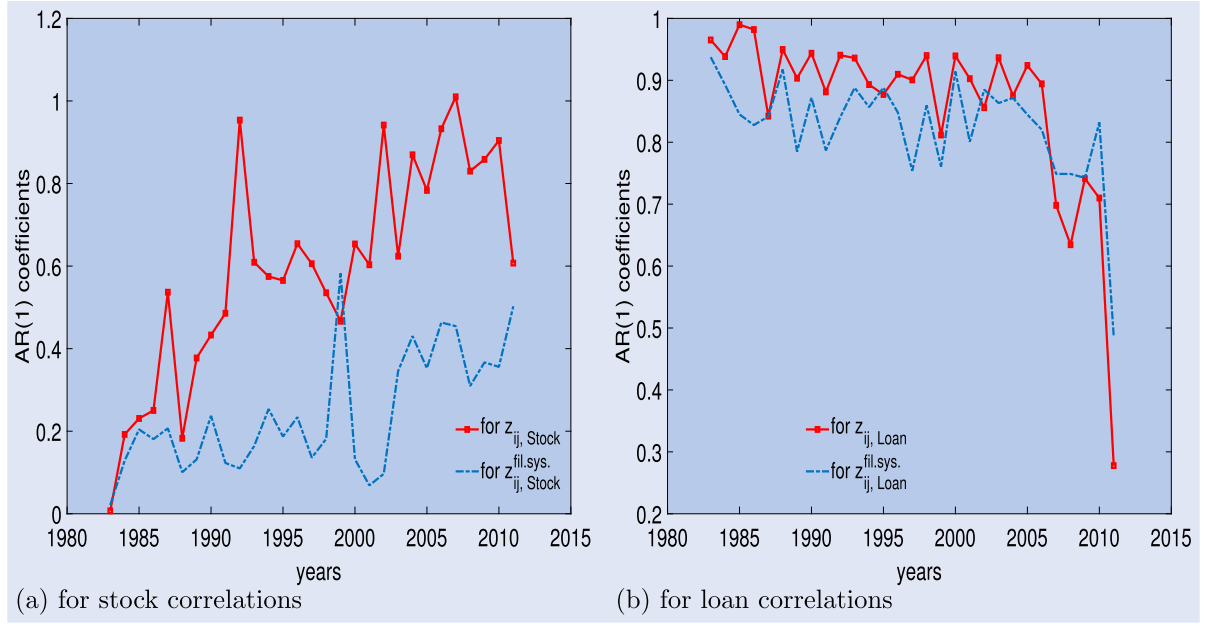


Figure 10. Degrees of correlation persistence based on the AR(1) coefficients, over the period from 1983 to 2011. Panels (a) and (b) show the results for stock return correlations and loan correlations, respectively. (a) for stock correlations and (b) for loan correlations.

Table 4. Model 1: Yearly similarities and loan raw correlations from 1983 to 1990.

Variables/Years	1983	1984	1985	1986	1987	1988	1989	1990
$z_{ij,Loan}(\tau - 1)$	0.9651*	0.9383*	0.9898*	0.9819*	0.8421*	0.9497*	0.9033*	0.9436*
	(192.2764)	(155.9339)	(171.3143)	(162.6154)	(111.8654)	(232.0672)	(164.7825)	(230.5395)
size sim.	-0.0003	0.0058**	0.0024	0.0057	0.0111*	0.0025	0.0007	0.0037
	(-0.1279)	(2.0225)	(0.8735)	(1.8876)	(2.8680)	(1.2855)	(0.2743)	(1.8769)
cash sim.	0.0002	0.0023	0.0011	-0.0128*	0.0191*	0.0036	0.0068**	0.0053*
	(0.0695)	(0.5391)	(0.2870)	(-3.1514)	(3.4560)	(1.3168)	(2.2389)	(2.6586)
security sim.	0.0075	-0.0223*	0.0071	-0.0302*	0.0005	-0.0029	0.0011	0.0089**
	(1.8705)	(-4.6860)	(1.4988)	(-5.1973)	(0.0705)	(-0.7743)	(0.2225)	(2.3696)
total loan sim.	-0.0021	-0.0276*	0.0176**	-0.0032	-0.0623*	-0.0034	-0.0068	0.0004
	(-0.3645)	(-3.0045)	(2.0745)	(-0.3037)	(-3.9681)	(-0.4191)	(-0.7169)	(0.0704)
deposit sim.	-0.0166*	-0.0149*	-0.0049	-0.0206*	-0.0124	-0.0068	-0.0135**	0.0005
	(-3.4811)	(-2.8397)	(-0.8970)	(-3.3053)	(-1.5165)	(-1.4702)	(-2.1838)	(0.1089)
equity sim.	0.0056	-0.0055	-0.0004	0.0082	0.0015	-0.0113**	-0.0311*	-0.0077
	(0.9218)	(-0.8173)	(-0.0607)	(1.2411)	(0.1799)	(-2.5289)	(-5.1780)	(-1.6797)
income sim.	-0.0108**	0.0066	-0.0058	-0.0155*	0.0166**	0.0002	0.0064	0.0144*
	(-2.1889)	(1.3185)	(-1.1644)	(-2.8194)	(2.5604)	(0.0731)	(1.2903)	(3.7284)
location sim.	0.0007	0.0104*	-0.0022	0.0042	-0.0033	-0.0025	0.0103*	0.0071*
	(0.4381)	(4.8090)	(-1.0099)	(1.7230)	(-1.0048)	(-1.4725)	(4.4624)	(4.2457)
type sim.	0.0024	0.0069**	-0.0079*	0.0006	0.0123*	0.0116*	0.0072*	-0.0033**
	(1.2442)	(2.5727)	(-2.9089)	(0.2239)	(3.2361)	(7.1845)	(3.3371)	(-2.1828)
observations	1892	2070	2256	2756	3080	4032	4290	4830
adj. R-squared	0.9661	0.9458	0.9476	0.9260	0.8445	0.9440	0.8862	0.9282

Notes: The dependent variable $z_{ij,Loan}(\tau)$ is the Fisher transformation of the Pearson correlation coefficient $C_{ij,Loan}(\tau)$. The model $z_{ij,Loan}(\tau) = \beta_0 z_{ij,Loan}(\tau - 1) + \sum_{k=1}^9 \beta_k \text{Similarity}_{ij}^k(\tau - 1) + \epsilon_{ij}(\tau)$ is estimated by OLS for each year τ . We report the coefficients for the lagged dependent variable as well as for the nine balance-sheet-based similarity variables in the table below with t-statistics in parentheses. * and ** indicate statistical significance at the 1% and 5% levels, respectively.

non-systematic characteristics of banks, and an idiosyncratic noise term as in the following

Model 2:

$$C_{ij}^{fil.sys.}(\tau) = f_2(C_{ij}^{fil.sys.}(\tau - 1), \text{Similarity}_{ij}^{(criterion_1)}(\tau - 1), \dots, \text{Similarity}_{ij}^{(criterion_n)}(\tau - 1)) + \epsilon_{ij}(\tau) \quad (20)$$

which we will once again operationalise using the same linear specification for ease of interpretability and comparability with the first model. One potential problem of the regression models in (19) and (20) is that while the correlation coefficients C_{ij} and $C_{ij}^{fil.sys.}$ are in $[-1, 1]$, the error terms are assumed to be distributed over the range $[-\infty, +\infty]$. To deal with this problem, following Pindyck and Rotemberg (1990)

Table 5. Model 1: Yearly similarities and loan raw correlations from 1991 to 2000.

	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
$z_{ij,Loan}(\tau - 1)$	0.8812*	0.9405*	0.9359*	0.8931*	0.8771*	0.9097*	0.9006*	0.9399*	0.8116*	0.9393*
	(168.8706)	(221.1929)	(239.6931)	(212.6035)	(225.6794)	(214.2535)	(151.0943)	(197.0614)	(120.1304)	(172.5413)
size sim.	-0.0017	0.0125*	0.0171*	0.0090*	0.0076*	0.0042*	0.0075*	-0.0051*	0.0039	-0.0028
	(-0.7976)	(7.2702)	(10.7007)	(5.5475)	(5.1202)	(2.7709)	(3.9657)	(-3.1701)	(1.6880)	(-1.5067)
cash sim.	0.0026	-0.0097*	0.0047*	0.0031	0.0031**	0.0061*	-0.0134*	-0.0060*	0.0053	0.0044
	(1.0670)	(-4.8360)	(2.5898)	(1.6721)	(1.9640)	(3.6251)	(-5.9120)	(-3.3918)	(1.9448)	(1.9458)
security sim.	-0.0044	0.0187*	0.0123*	0.0142*	0.0040	0.0099*	0.0092	0.0133*	0.0621*	0.0014
	(-0.9020)	(4.6177)	(3.6696)	(3.7179)	(1.1927)	(2.6914)	(1.9434)	(3.9455)	(11.6860)	(0.3023)
total loan sim.	0.0143	0.0247*	0.0025	-0.0424*	-0.0050	-0.0051	0.0474*	-0.0073	-0.0101	-0.0114
	(1.8549)	(3.9738)	(0.3368)	(-5.3979)	(-0.6450)	(-0.5999)	(3.9586)	(-0.8022)	(-0.7705)	(-0.9004)
deposit sim.	0.0023	-0.0005	-0.0082**	0.0117*	-0.0093*	0.0013	-0.0383*	-0.0067	0.0040	-0.0440*
	(0.3825)	(-0.1340)	(-2.2953)	(3.3845)	(-3.1762)	(0.4523)	(-10.0166)	(-1.7602)	(0.7568)	(-3.9264)
equity sim.	0.0093	-0.0047	0.0138*	0.0104**	-0.0068	0.0060	0.0007	-0.0013	-0.0178*	0.0045
	(1.3725)	(-0.8754)	(2.8915)	(2.1410)	(-1.5865)	(1.5377)	(0.1607)	(-0.4061)	(-4.4843)	(1.7956)
income sim.	-0.0033	0.0068**	-0.0000	0.0011	0.0178*	0.0000	0.0021	0.0000*	0.0000	-0.0014**
	(-0.6206)	(2.5285)	(-0.1293)	(0.6779)	(15.8188)	(0.8503)	(1.8875)	(4.4053)	(0.6727)	(-2.1077)
location sim.	0.0202*	0.0079*	0.0062*	0.0161*	0.0140*	0.0183*	0.0054**	0.0055*	0.0302*	0.0139*
	(8.7953)	(4.2904)	(3.6099)	(9.0688)	(8.8277)	(10.5812)	(2.2946)	(2.8841)	(10.6577)	(3.8860)
type sim.	-0.0061*	-0.0096*	-0.0088*	-0.0045*	-0.0026**	-0.0034**	0.0022	0.0060*	-0.0038	0.0297*
	(-3.1764)	(-6.2703)	(-6.5389)	(-3.2370)	(-2.1904)	(-2.5329)	(1.5076)	(3.8515)	(-1.7916)	(8.8292)
observations	5402	5550	6162	6162	5700	5550	5852	5852	5402	4422
adj. R-squared	0.8617	0.9128	0.9186	0.9036	0.9242	0.9137	0.8270	0.8963	0.7812	0.8904

Notes: The dependent variable $z_{ij,Loan}(\tau)$ is the Fisher transformation of the Pearson correlation coefficient $C_{ij,Loan}(\tau)$. The model $z_{ij,Loan}(\tau) = \beta_0 z_{ij,Loan}(\tau - 1) + \sum_{k=1}^9 \beta_k \text{Similarity}_{ij}^k(\tau - 1) + \epsilon_{ij}(\tau)$ is estimated by OLS for each year τ . We report the coefficients for the lagged dependent variable as well as for the nine balance-sheet-based similarity variables in the table below with t-statistics in parentheses. * and ** indicate statistical significance at the 1% and 5% levels, respectively.

and Box (2018) we use the Fisher transformation[†] for the correlation coefficients $C_{ij}(\tau)$ as

$$z_{ij}(\tau) = \frac{1}{2} \ln \frac{1 + C_{ij}(\tau)}{1 - C_{ij}(\tau)} \quad (21)$$

and, similarly, for $C_{ij}^{fil.sys.}(\tau)$ as

$$z_{ij}^{fil.sys.}(\tau) = \frac{1}{2} \ln \frac{1 + C_{ij}^{fil.sys.}(\tau)}{1 - C_{ij}^{fil.sys.}(\tau)}. \quad (22)$$

Once obtained, these Fisher transformation coefficients will replace C_{ij} and $C_{ij}^{fil.sys.}$ in (19) and (20), respectively.

4.2. Estimation results

We first investigate how raw correlations $\{C_{ij,Stock}\}$ or $\{C_{ij,Loan}\}$ are affected by the one-year lag correlations and the qualitative and quantitative similarities between banks. To this end, using the OLS method, we estimate the coefficients of the first model shown in equation (19).

Tables 1, 2 and 3 report estimates of equation (19) for the raw correlations $\{C_{ij,Stock}\}$ of stock returns over the period from 1983 to 2011. The results in the first row of these tables show a certain persistence of pairwise correlations: For almost all of the years, high correlations in the previous year tend to result in large correlations in the present year. Among the quantitative similarities, we find that the coefficient of

size similarity is often positive and significant. The remaining control variables are also statistically significant at the 1% or 5% levels in many years. Nevertheless, the signs of their coefficients change more often.[‡] Overall, for most of the years from 1991 to 2010, the adjusted R-squared implies that more than 45% of the variance of yearly stock return correlation is accounted for by the lag of the dependent variable and the quantitative as well as qualitative similarities of the characteristics of banks.

Regarding the credit market, the estimation results for loan correlations $\{C_{ij,Loan}\}$ over the sample period are reported in tables 4, 5 and 6. Similarly, the loan correlation in the previous year always acts as an important predictor. For a further detailed comparison between the two layers, in figure 10 we plot the evolution of the estimated coefficients of $\{z_{ij,Stock}(\tau - 1)\}$ and $\{z_{ij,Loan}(\tau - 1)\}$ over the years. As shown in the red dotted curves in Panels (a) and (b), the raw correlations of stock returns and those of loan correlations exhibit different levels of persistence in different periods. Before the time of the financial crisis, while the stock market correlations are characterised by volatile periods, the loan correlations behave more sluggishly. The coefficient of $\{z_{ij,Stock}(\tau - 1)\}$ reaches the first peak at 0.9535 after the collapse of the Japanese asset price bubble and the second peak at 1.009 corresponding to the time of the global financial crisis, but we do not observe such a high level of dependencies of the loan correlation on its one-year lag. However, for both markets, the coefficients of lagged dependent variables significantly decrease in 2011–the year when the country was hit by the Tōhoku earthquake

[†] Theoretically, this transformed correlation coefficient has the range $[-\infty, +\infty]$.

[‡] For brevity, they are not reported in detail, but a comprehensive set of results is documented in tables 1 to 3.

Table 6. Model 1: Yearly similarities and *loan raw correlations* from 2001 to 2011.

Variables/Years	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
$z_{ij,Loan}(\tau - 1)$	0.9026*	0.8557*	0.9364*	0.8750*	0.9239*	0.8942*	0.6980*	0.6344*	0.7412*	0.7098*	0.2777*
	(127.8436)	(122.5153)	(155.5785)	(142.0162)	(155.4340)	(129.1927)	(61.0978)	(66.8274)	(53.0574)	(109.4716)	(41.1688)
size sim.	0.0077*	0.0073*	−0.0015	0.0028	0.0007	0.0111*	0.0080**	0.0225*	0.0176*	0.0170*	0.0392*
	(3.0691)	(2.9387)	(−0.7558)	(1.3236)	(0.3687)	(5.1567)	(2.2075)	(7.3060)	(4.3826)	(6.4053)	(8.5241)
cash sim.	0.0035	0.0060**	−0.0015	−0.0078*	−0.0013	−0.0016	−0.0503*	−0.0130*	−0.0252*	−0.0101*	0.0033
	(1.2004)	(2.0651)	(−0.6273)	(−3.3455)	(−0.5856)	(−0.6500)	(−10.9533)	(−3.1771)	(−5.0229)	(−2.9362)	(0.5780)
security sim.	0.0113	0.0378*	0.0308*	0.0062	0.0261*	0.0067	−0.0010	0.0192*	0.0496*	0.0042	0.0279**
	(1.8591)	(6.2661)	(7.6126)	(1.4760)	(6.1253)	(1.2704)	(−0.1043)	(2.6431)	(5.0718)	(0.6029)	(2.2660)
total loan sim.	−0.0190	−0.0160	−0.0559*	−0.0000	−0.0083	0.0124	0.0476**	−0.0167	−0.0971*	−0.0430*	0.0549
	(−1.1440)	(−0.8786)	(−4.4337)	(−0.0055)	(−0.7510)	(0.8824)	(2.0374)	(−0.9339)	(−3.7491)	(−2.6857)	(1.7536)
deposit sim.	0.0166	0.2056*	−0.0258	−0.0052	−0.0068	0.0055	0.0044	−0.0341*	0.0728*	0.0084	−0.0135
	(0.4955)	(5.2808)	(−1.7382)	(−0.6557)	(−0.8260)	(0.6373)	(0.3361)	(−3.1669)	(4.4852)	(0.7642)	(−0.5894)
equity sim.	−0.0062	−0.0245*	−0.0016	−0.0059	−0.0004	0.0214*	−0.0141**	0.0175*	−0.0491*	0.0131**	0.0573
	(−0.9820)	(−4.5032)	(−0.5149)	(−1.3951)	(−0.1015)	(5.9104)	(−1.9627)	(2.8836)	(−6.5956)	(2.1537)	(5.7174)
income sim.	−0.0000	−0.0000	−0.0002	−0.0000	0.0001*	−0.0000	−0.0002	0.0001**	0.0000	0.0000	0.0000
	(−0.4034)	(−0.4369)	(−1.0337)	(−0.0007)	(2.9179)	(−1.3655)	(−1.4947)	(2.0668)	(1.6962)	(0.2722)	(0.1072)
location sim.	0.0237*	0.0505*	−0.0003	0.0248*	0.0057	0.0350*	0.0441*	0.0926*	0.0327*	0.0659*	0.1958*
	(5.0071)	(10.9033)	(−0.0936)	(6.7318)	(1.6962)	(8.7330)	(6.8309)	(17.8784)	(4.5349)	(13.7519)	(22.3337)
type sim.	−0.0092	−0.0711*	0.0149*	0.0023	0.0029	0.0045	0.0186*	−0.0138*	0.0199*	−0.0042	0.0156*
	(−0.7578)	(−5.7904)	(4.3260)	(1.0317)	(1.2712)	(1.8025)	(4.7232)	(−4.3366)	(4.7636)	(−1.5628)	(3.0048)
observations	4032	3906	4290	4556	4422	4830	4970	4830	4692	5112	4830
adj. R-squared	0.8381	0.8403	0.8773	0.8495	0.8741	0.8200	0.5181	0.5906	0.4810	0.7419	0.3733

Notes: The dependent variable $z_{ij,Loan}(\tau)$ is the Fisher transformation of the Pearson correlation coefficient $C_{ij,Loan}(\tau)$. The model $z_{ij,Loan}(\tau) = \beta_0 z_{ij,Loan}(\tau - 1) + \sum_{k=1}^9 \beta_k \text{Similarity}_{ij}^k(\tau - 1) + \epsilon_{ij}(\tau)$ is estimated by OLS for each year τ . We report the coefficients for the lagged dependent variable as well as for the nine balance-sheet-based similarity variables in the table below with t-statistics in parentheses. * and ** indicate statistical significance at the 1% and 5% levels, respectively.

Table 7. Model 2: Yearly similarities and *filtered stock return correlations* from 1983 to 1990.

Variables/Years	1983	1984	1985	1986	1987	1988	1989	1990
$z_{ij,Stock}^{fil.sys.}(\tau - 1)$	0.0214 (0.5468)	0.1286* (8.7835)	0.2043* (12.2137)	0.1808* (11.3724)	0.2068* (13.0597)	0.1006* (10.5204)	0.1316* (7.1077)	0.2373* (24.7084)
size sim.	0.0539* (2.9083)	0.1719* (13.9363)	0.0945* (9.2448)	0.0718* (8.9962)	0.1313* (18.4311)	0.0528* (12.7369)	0.0490* (10.2992)	0.0761* (24.5193)
cash sim.	0.0503 (1.8540)	0.0029 (0.1513)	-0.0081 (-0.5509)	0.0164 (1.5427)	0.0768* (7.5999)	0.0268* (4.5422)	0.0493* (9.1045)	0.0522* (16.6058)
security sim.	0.0542 (1.6644)	0.0969* (4.4615)	-0.0213 (-1.2045)	0.1069* (7.0447)	-0.0488* (-3.4227)	0.0047 (0.5866)	-0.0103 (-1.1029)	-0.0213* (-3.6856)
total loan sim.	-0.1218** (-2.5104)	0.2750* (6.5824)	0.1091* (3.4601)	0.0601** (2.1477)	-0.0137 (-0.4781)	0.0442** (2.5572)	0.0811* (4.7999)	0.0225** (2.3891)
deposit sim.	-0.0978** (-2.5137)	0.0842* (3.5191)	-0.0831* (-4.0549)	0.0587* (3.6216)	-0.0257 (-1.7120)	0.0719* (7.4220)	-0.0080 (-0.7335)	0.0183** (2.5018)
equity sim.	0.2073* (4.2054)	-0.1152* (-3.7504)	-0.0903* (-3.6799)	0.0571* (3.2675)	-0.0287 (-1.8391)	-0.0115 (-1.2379)	0.0073 (0.6948)	-0.0142** (-1.9985)
income sim.	-0.1931* (-4.9090)	0.0243 (1.0666)	0.0046 (0.2521)	-0.0055 (-0.3834)	0.0025 (0.2132)	0.0284* (4.2141)	-0.0372* (-4.2017)	-0.0141** (-2.3589)
location sim.	0.0984* (7.4419)	0.0998* (10.2754)	0.0732* (9.0728)	0.1499* (23.1063)	0.0261* (3.9602)	0.0397* (11.4074)	0.0853* (21.3732)	0.0697* (26.4141)
type sim.	0.1046* (6.5527)	0.0238 (1.9353)	0.0254** (2.5457)	0.0004 (0.0558)	0.0234* (3.3802)	0.0032 (0.9520)	-0.0119* (-3.1389)	-0.0023 (-0.9842)
observations	1892	2070	2256	2756	3080	4032	4290	4830
adj. R-squared	0.2758	0.4769	0.3419	0.4999	0.5638	0.4341	0.2992	0.6070

Notes: The dependent variable $z_{ij,Stock}^{fil.sys.}(\tau)$ is the Fisher transformation of the Pearson correlation coefficient $C_{ij,Stock}^{fil.sys.}(\tau)$. The model $z_{ij,Stock}^{fil.sys.}(\tau) = \beta_0 z_{ij,Stock}^{fil.sys.}(\tau - 1) + \sum_{k=1}^9 \beta_k \text{Similarity}_{ij}^k(\tau - 1) + \epsilon_{ij}(\tau)$ is estimated by OLS for each year τ . We report the coefficients for the lagged dependent variable as well as for the nine balance-sheet-based similarity variables in the table below with t-statistics in parentheses. * and ** indicate statistical significance at the 1% and 5% levels, respectively.

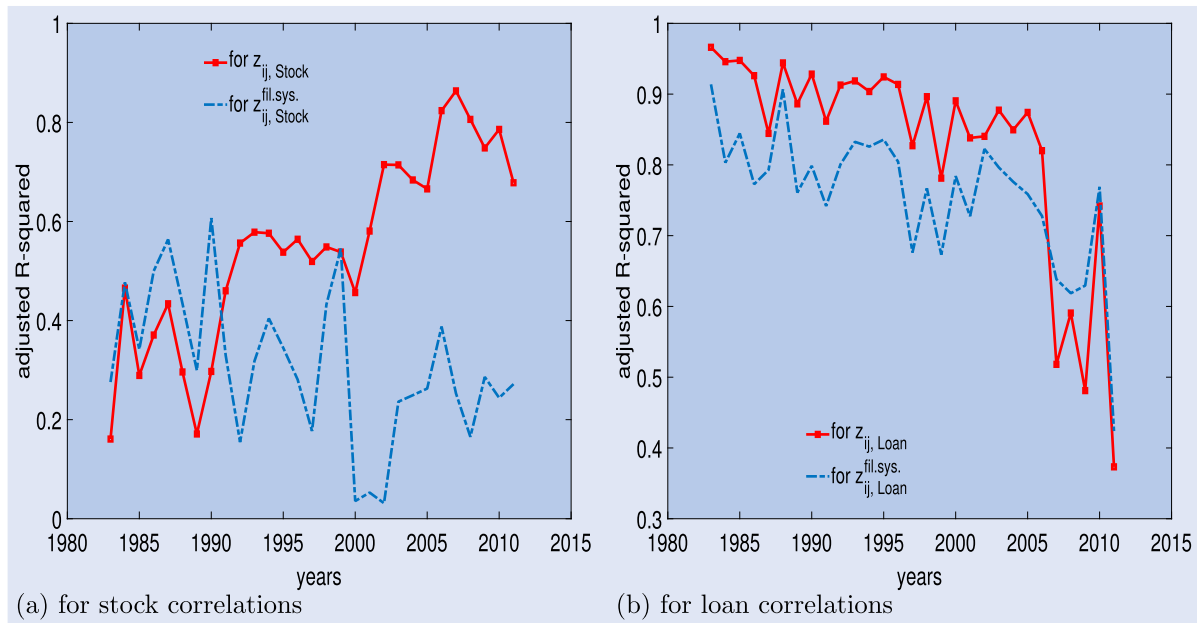


Figure 11. The values of the adjusted R squared in the two models for raw correlations and filtered correlations over the period from 1983 to 2011. (a) for stock correlations and (b) for loan correlations.

and tsunami. A similar observation can be obtained from the changes in the values of adjusted R-squared. This may indicate a structural break of correlations among Japanese banks in *both markets* under the severe effects of such a natural disaster.

As explained in Section 2.2.3, the interdependencies among banks can be driven by different components, and the common market mode may play a prominent role. To examine how much of the residual correlations can be accounted for by the non-systematic components such as those related to the similar characteristics and fundamental information of banks,

Table 8. Model 2: Yearly similarities and *filtered stock return correlations* from 1991 to 2000.

	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
$z_{ij,Stock}^{fil.sys.}(\tau - 1)$	0.1229* (9.6548)	0.1098* (5.6899)	0.1642* (14.6131)	0.2538* (19.6035)	0.1871* (15.4943)	0.2340* (14.1813)	0.1359* (9.7321)	0.1813* (17.5636)	0.5823* (39.8312)	0.1320* (8.157)
size sim.	0.0308* (12.0693)	0.0420* (11.9295)	0.0451* (15.2155)	0.0347* (11.2804)	0.0330* (10.8823)	0.0460* (12.6)	0.0422* (11.5007)	0.0365* (12.0208)	0.0443* (12.6326)	0.0320* (8.1834)
cash sim.	0.0359* (12.6477)	0.0038 (0.9463)	0.0235* (6.9673)	0.0109* (3.1236)	0.0184* (5.5600)	0.0302* (7.4013)	0.0140* (3.2302)	0.0051 (1.5302)	0.0182* (4.4445)	− 0.0017 (− 0.3652)
security sim.	0.0157* (2.9511)	0.0377* (4.6561)	0.0259* (4.2354)	− 0.0333* (− 4.6548)	− 0.0043 (− 0.6214)	0.0347* (3.9126)	0.0247* (2.721)	− 0.0045 (− 0.7230)	0.0054 (0.6899)	− 0.0298* (− 2.9902)
total loan sim.	− 0.0387* (− 4.6341)	− 0.0246** (− 1.9747)	− 0.0777* (− 5.6451)	− 0.0274 (− 1.8678)	− 0.0519* (− 3.2753)	− 0.0486** (− 2.3341)	− 0.0066 (− 0.2893)	− 0.0492* (− 2.8505)	0.0479** (2.4341)	9.6303e-05 (0.0036)
deposit sim.	0.0270* (4.0882)	− 0.0208** (− 2.3593)	− 0.0044 (− 0.6754)	0.0271* (4.1841)	0.0141** (2.3590)	0.0152** (2.0824)	0.0403* (5.4779)	0.0122 (1.6819)	− 0.0017 (− 0.2109)	0.0326 (1.4120)
equity sim.	− 0.0061 (− 0.8438)	0.0027 (0.2566)	0.0597* (6.7616)	− 0.0659* (− 7.1916)	0.0247* (2.8230)	0.0082 (0.8763)	0.0627* (6.9136)	0.0737* (12.0795)	0.0155* (2.6301)	0.0034 (0.6645)
income sim.	0.0050 (0.8795)	− 0.0118** (− 2.1982)	− 0.0001 (− 1.5914)	0.0331* (10.7452)	0.0040 (1.8039)	− 0.0000 (− 0.1413)	0.0019 (0.8948)	0.0000 (1.7533)	0.0004** (2.4106)	− 0.0011 (− 0.8595)
location sim.	0.0177* (6.6691)	0.0248* (6.9224)	0.0547* (17.7334)	0.0459* (13.8448)	0.0713* (22.3502)	0.0656* (15.6408)	0.0595* (13.3535)	0.1154* (33.0547)	0.0855* (19.5008)	0.0049 (0.7116)
type sim.	0.0082* (3.9301)	0.0170* (5.5596)	0.0263* (10.4270)	0.0406* (15.5075)	0.0201* (7.9816)	0.0108* (3.2663)	0.0057** (2.0452)	0.0357* (12.1069)	0.0249* (7.578)	− 0.0354* (− 5.0818)
observations	5402	5550	6162	6162	5700	5550	5852	5852	5402	4422
adj. R-squared	0.3299	0.1546	0.3182	0.4043	0.3449	0.2809	0.1770	0.4325	0.5452	0.0356

Notes: The dependent variable $z_{ij,Stock}^{fil.sys.}(\tau)$ is the Fisher transformation of the Pearson correlation coefficient $C_{ij,Stock}^{fil.sys.}(\tau)$. The model $z_{ij,Stock}^{fil.sys.}(\tau) = \beta_0 z_{ij,Stock}^{fil.sys.}(\tau - 1) + \sum_{k=1}^9 \beta_k \text{Similarity}_{ij}^k(\tau - 1) + \epsilon_{ij}(\tau)$ is estimated by OLS for each year τ . We report the coefficients for the lagged dependent variable as well as for the nine balance-sheet-based similarity variables in the table below with t-statistics in parentheses. * and ** indicate statistical significance at the 1% and 5% levels, respectively.

we first need to filter out the effects of the market mode to obtain the filtered correlations $\{C_{ij}^{fil.sys.}\}$ and estimate the coefficients of the second model shown in equation (20) for the different years from 1983 to 2011.

We report the estimation results for the filtered stock return correlations $C_{ij,Stock}^{fil.sys.}$ in tables 7, 8 and 9. Among the different explanatory variables, the one-year lagged value of the dependent variable and the size similarity are still positive and significant for most of the years. However, in general, the value of the coefficient of $z_{ij,Stock}^{fil.sys.}$ is significantly reduced when we subtract the effects of the market mode. Another interesting distinction is the different values of the adjusted R-squared statistics in the two models (see also the panel (a) of figure 11). Intuitively, such a difference can be related to the contribution of different components to the correlations stemming from the equity market: When the contribution of the market mode to the total variance of stock returns becomes relatively weaker and, thus, other non-systematic factors such as banks' characteristics play a more important role, excluding the effects of the market mode typically results in a better goodness of fit (consult also figures 8 and A5 in the Appendix).

Moving onto the filtered correlations $C_{ij,Loan}^{fil.sys.}$ for loan portfolios, the differences from the estimation results of the first model are less pronounced. As summarised in tables 10, 11 and 12, up to 2010, the coefficient of the previous filtered correlation stays positive (mostly around 0.7 to 0.9) and statistically significant, and then sharply decreases (but remains positive) in the aftermath of the 2011 Tōhoku earthquake and tsunami. Among the other explanatory variables, size similarities and location are also significant over almost all of the years. Notice that the adjusted R-squared only slightly

changed after the effects of the common market mode are filtered out in the credit market (see also the panel (b) of figure 11). This may be because, first, unlike in the equity market, there are more significant components driving the loan correlation dynamics in the credit market (see figure A5 in the Appendix). Second, among these components, the proportion of the dominant market mode contributing to the total variance in loan correlations is relatively smaller than its counterpart in stock return correlations.[†] Therefore, it is unsurprising if the adjusted R-squared does not change significantly when we exclude the market mode in the credit market.

All in all, we find that both raw and filtered correlations exhibit a certain degree of persistence and that no other explanatory variables based on quantitative or qualitative similarities can predict future comovements better than contemporaneous comovements. Furthermore, in line with the results obtained from the principal component analysis in section (3), on top of having distinct structures over the years, the correlations stemming from two market layers also respond differently to the external shocks such as the collapse of the asset price bubble in the Japanese economy, the 2007-2008 crash of the global financial system, or the 2011 natural disaster.

[†] As depicted in figures 7 and 8, the value of AR₁ extracted from the credit market layer is smaller than that from the equity market layer.

Table 9. Model 2: Yearly similarities and *filtered stock return correlations* from 2001 to 2011.

Variables/Years	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
$z_{ij,Stock}^{fil.sys.}(\tau - 1)$	0.0687*	0.0967*	0.3457*	0.4303*	0.3525*	0.4634*	0.4543*	0.3095*	0.3668*	0.3557*	0.5025*
	(6.0670)	(4.6486)	(22.2767)	(30.6325)	(32.5767)	(48.1016)	(33.8110)	(20.6236)	(26.3180)	(28.7217)	(31.3571)
size sim.	0.0364*	0.0241*	0.0791*	0.0216*	0.0201*	0.0248*	0.0169*	0.0323*	0.0433*	0.0345*	0.0427*
	(12.3049)	(6.2342)	(19.0099)	(4.8790)	(6.0728)	(10.7812)	(6.3210)	(10.3144)	(14.5299)	(12.3951)	(12.6021)
cash sim.	-0.0011	0.0093**	0.0009	-0.0087	0.0094**	-0.0035	0.0019	0.0017	0.0103*	0.0051	-0.0042
	(-0.3265)	(2.0920)	(0.1990)	(-1.8663)	(2.3753)	(-1.2590)	(0.5959)	(0.4259)	(2.7759)	(1.4558)	(-1.0310)
security sim.	-0.0170**	-0.0231**	-0.0265*	0.0219*	0.0247*	0.0071	0.0119	0.0235*	-0.0055	0.0092	-0.0217**
	(-2.4184)	(-2.5125)	(-3.2582)	(2.6158)	(3.4956)	(1.2638)	(1.6968)	(3.1986)	(-0.7657)	(1.3022)	(-2.4724)
total loan sim.	-0.0061	0.0275	0.0305	-0.0570**	-0.0727*	0.0268	-0.0580*	-0.0574*	-0.0022	-0.0558*	0.0203
	(-0.3166)	(0.9793)	(1.1975)	(-2.3516)	(-3.8865)	(1.7783)	(-3.4150)	(-3.1544)	(-0.1189)	(-3.3915)	(0.9131)
deposit sim.	-0.0829**	0.0372	0.1993*	0.0523*	0.0058	0.0162	0.0076	0.0147	-0.0377*	0.0208	-0.0461*
	(-2.1421)	(0.6303)	(6.6283)	(3.2419)	(0.4227)	(1.7490)	(0.7956)	(1.3526)	(-3.1385)	(1.8379)	(-2.8147)
equity sim.	0.0063	0.0172**	-0.0016	0.0695*	-0.0020	0.0105*	0.0039	0.0283*	0.0488*	0.0382*	0.0726*
	(0.8562)	(2.0663)	(-0.2481)	(8.1891)	(-0.3050)	(2.7337)	(0.7475)	(4.5724)	(8.7715)	(6.0460)	(10.1053)
income sim.	0.0000	-0.0001	0.0012*	0.0000	-0.0001**	0.0001*	-0.0000	-0.0000	0.0000	-0.0000	0.0000**
	(0.5212)	(-1.0289)	(3.0907)	(0.1750)	(-2.1770)	(2.7479)	(-0.5374)	(-0.1462)	(0.0790)	(-1.4759)	(2.5438)
location sim.	0.0135*	0.0217*	0.0745*	0.0267*	0.0453*	0.0216*	0.0405*	0.0452*	0.0736*	0.0464*	0.0390*
	(2.6561)	(3.2953)	(10.6399)	(3.7151)	(8.4735)	(5.3299)	(9.1641)	(9.0461)	(14.8890)	(9.5329)	(6.2653)
type sim.	0.0251	4.4054e-05	-0.0521*	-0.0072	0.0026	-0.0076*	0.0024	-0.0034	0.0109*	-0.0063**	-0.0041
	(1.7822)	(0.0023)	(-7.4435)	(-1.6138)	(0.6923)	(-2.8607)	(0.8416)	(-1.0608)	(3.5363)	(-2.3041)	(-1.1332)
observations	4032	3906	4290	4556	4422	4830	4970	4830	4692	5112	4830
adj. R-squared	0.0528	0.0314	0.2362	0.2493	0.2627	0.3878	0.2530	0.1653	0.2862	0.2440	0.2717

Notes: The dependent variable $z_{ij,Stock}^{fil.sys.}(\tau)$ is the Fisher transformation of the Pearson correlation coefficient $C_{ij,Stock}^{fil.sys.}(\tau)$. The model $z_{ij,Stock}^{fil.sys.}(\tau) = \beta_0 z_{ij,Stock}^{fil.sys.}(\tau - 1) + \sum_{k=1}^9 \beta_k \text{Similarity}_{ij}^k(\tau - 1) + \epsilon_{ij}(\tau)$ is estimated by OLS for each year τ . We report the coefficients for the lagged dependent variable as well as for the nine balance-sheet-based similarity variables in the table below with t-statistics in parentheses. * and ** indicate statistical significance at the 1% and 5% levels, respectively

Table 10. Model 2: Yearly similarities and *filtered loan correlations* from 1983 to 1990.

Variables/Years	1983	1984	1985	1986	1987	1988	1989	1990
$z_{ij,Loan}^{fil.sys.}(\tau - 1)$	0.9373*	0.8926*	0.8450*	0.8276*	0.8416*	0.9175*	0.7849*	0.8718*
	(114.0595)	(71.8004)	(89.8576)	(79.2599)	(89.2315)	(168.0031)	(94.6778)	(116.4721)
size sim.	0.0046	0.0058	-0.0119*	0.0132*	0.0230*	0.0076*	0.0232*	0.0053
	(1.2436)	(1.1501)	(-2.7839)	(2.7936)	(5.3730)	(3.0877)	(6.5002)	(1.7524)
cash sim.	-0.0155*	0.0086	0.0193*	0.0128	-0.0057	0.0064	0.0031	0.00124
	(-2.8711)	(1.0644)	(2.9578)	(1.9484)	(-0.9158)	(1.8340)	(0.7687)	(0.4062)
security sim.	-0.0005	0.0021	0.0158**	0.0018	0.0002	-0.0040	-0.0125	0.0333*
	(-0.0892)	(0.2374)	(2.0590)	(0.2020)	(0.0328)	(-0.8305)	(-1.7590)	(5.8112)
total loan sim.	0.0211**	0.0079	-0.0085	-0.0052	0.1062*	0.0188	-0.0022	0.0312*
	(2.2318)	(0.4647)	(-0.6205)	(-0.3110)	(6.0097)	(1.8013)	(-0.1743)	(3.3592)
deposit sim.	0.0121	-0.0023	0.0142	-0.0128	-0.0145	0.0046	-0.0098	-0.0041
	(1.5675)	(-0.2394)	(1.5836)	(-1.2913)	(-1.5778)	(0.7924)	(-1.1869)	(-0.5721)
equity sim.	0.0026	-0.0208	0.0000	0.0147	-0.0065	0.0018	0.0091	0.0017
	(0.2719)	(-1.6495)	(0.0062)	(1.3850)	(-0.6779)	(0.3333)	(1.1357)	(0.2417)
income sim.	0.0131	0.0231**	-0.0088	0.0048	-0.0221*	-0.0086**	-0.0160**	0.0151**
	(1.6794)	(2.4590)	(-1.0943)	(0.5520)	(-3.0285)	(-2.1071)	(-2.3997)	(2.5658)
location sim.	0.0084*	0.0116*	0.0254*	0.0130*	0.0147*	0.0170*	0.0330*	0.0243*
	(3.0489)	(2.7988)	(7.1134)	(3.2390)	(3.8602)	(7.9560)	(10.4955)	(9.1453)
type sim.	0.0003	0.0037	-0.0052	-0.0075	-0.0042	-0.0053*	-0.0026	-0.0082*
	(0.1203)	(0.7526)	(-1.2039)	(-1.6879)	(-1.0030)	(-2.6241)	(-0.9342)	(-3.5889)
observations	1892	2070	2256	2756	3080	4032	4290	4830
adj. R-squared	0.9137	0.8031	0.8444	0.7726	0.7928	0.9066	0.7608	0.7986

Notes: The dependent variable $z_{ij,Loan}^{fil.sys.}(\tau)$ is the Fisher transformation of the Pearson correlation coefficient $C_{ij,Loan}^{fil.sys.}(\tau)$. The model $z_{ij,Loan}^{fil.sys.}(\tau) = \beta_0 z_{ij,Loan}^{fil.sys.}(\tau - 1) + \sum_{k=1}^9 \beta_k \text{Similarity}_{ij}^k(\tau - 1) + \epsilon_{ij}(\tau)$ is estimated by OLS for each year τ . We report the coefficients for the lagged dependent variable as well as for the nine balance-sheet-based similarity variables in the table below with t-statistics in parentheses. * and ** indicate statistical significance at the 1% and 5% levels, respectively.

Table 11. Model 2: Yearly similarities and *filtered loan correlations* from 1991 to 2000.

	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
$z_{ij,Loan}^{fil.sys.}(\tau - 1)$	0.7867*	0.8404*	0.8879*	0.8566*	0.8881*	0.8481*	0.7537*	0.8595*	0.7608*	0.9142*
	(105.1760)	(128.0602)	(151.5319)	(147.2161)	(148.2592)	(127.5148)	(99.5270)	(121.9839)	(89.6314)	(115.2931)
size sim.	0.0079*	0.0148*	0.0084*	0.0100*	-0.0045**	0.0101*	-0.0004	0.0023	0.0070**	0.0010
	(2.7611)	(6.1295)	(3.8545)	(4.6992)	(-2.0625)	(4.1813)	(-0.1760)	(0.9714)	(2.3194)	(0.3576)
cash sim.	0.0016	-0.0014	0.0065*	0.0050**	0.0028	0.0020	-0.0017	-0.0008	0.0007	-0.0062
	(0.5001)	(-0.5054)	(2.6255)	(2.0616)	(1.1698)	(0.7593)	(-0.5687)	(-0.3347)	(0.1957)	(-1.7476)
security sim.	0.0095	0.0049	0.0138*	0.0030	0.0021	0.0066	0.0099	0.0036	-0.0030	-0.0001
	(1.5193)	(0.8779)	(3.0774)	(0.6061)	(0.4263)	(1.1235)	(1.5497)	(0.7221)	(-0.4403)	(-0.0147)
total loan sim.	0.0153	0.0233*	0.0130	-0.0109	-0.0145	0.0039	0.0131	0.0115	0.0295	0.0457**
	(1.5383)	(2.6719)	(1.2902)	(-1.0659)	(-1.2488)	(0.2848)	(0.8126)	(0.8372)	(1.7163)	(2.3103)
deposit sim.	0.0006	-0.0004	-0.0072	-0.0003	0.0083	0.0051	-0.0062	-0.0024	0.0051	-0.0121
	(0.0767)	(-0.0699)	(-1.4918)	(-0.0774)	(1.8959)	(1.0705)	(-1.2151)	(-0.4314)	(0.7354)	(-0.6924)
equity sim.	0.0019	0.0034	-0.0129**	0.0197*	0.0061	-0.0123**	-0.0001	-0.0011	-0.0067	-0.0040
	(0.2219)	(0.4617)	(-2.0045)	(3.1010)	(0.9565)	(-1.9831)	(-0.0246)	(-0.2322)	(-1.3100)	(-1.0289)
income sim.	-0.0014	-0.0068	-0.0000	0.0051**	0.0015	0.0000	0.0020	0.0000	0.0001	0.0012
	(-0.2084)	(-1.8291)	(-0.4793)	(2.3911)	(0.9773)	(0.7786)	(1.3524)	(0.0494)	(0.9980)	(1.2023)
location sim.	0.0280*	0.0204*	0.0119*	0.0204*	0.0267*	0.0303*	0.0245*	0.0230*	0.0555*	0.0409*
	(9.2667)	(7.7950)	(5.0090)	(8.5890)	(11.0796)	(10.8021)	(7.6495)	(7.9165)	(14.8761)	(7.3345)
type sim.	0.0005	-0.0054**	0.0001	0.0001	0.0009	0.0030	0.0040**	0.0009	-0.0048	-0.0206*
	(0.2171)	(-2.5481)	(0.0830)	(0.0624)	(0.5494)	(1.3996)	(2.0438)	(0.3943)	(-1.7190)	(-3.9402)
observations	5402	5550	6162	6162	5700	5550	5852	5852	5402	4422
adj. R-squared	0.7418	0.8007	0.8325	0.8258	0.8357	0.8044	0.6761	0.7669	0.6730	0.7841

Notes: The dependent variable $z_{ij,Loan}^{fil.sys.}(\tau)$ is the Fisher transformation of the Pearson correlation coefficient $C_{ij,Loan}^{fil.sys.}(\tau)$. The model $z_{ij,Loan}^{fil.sys.}(\tau) = \beta_0 z_{ij,Loan}^{fil.sys.}(\tau - 1) + \sum_{k=1}^9 \beta_k \text{Similarity}_{ij}^k(\tau - 1) + \epsilon_{ij}(\tau)$ is estimated by OLS for each year τ . We report the coefficients for the lagged dependent variable as well as for the nine balance-sheet-based similarity variables in the table below with t-statistics in parentheses. * and ** indicate statistical significance at the 1% and 5% levels, respectively.

5. Discussions and concluding remarks

We have analysed the multilayer nature of interactions in the banking system of Japan, focussing on loan portfolio

correlations arising from the credit market and stock return correlations stemming from the equity market. Using a unified procedure based on the methods of random matrix theory, principal component analysis and network approaches, we

Table 12. Model 2: Yearly similarities and *filtered loan correlations* from 2001 to 2011.

Variables/Years	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
$z_{ij,Loan}^{fil.sys.}(\tau - 1)$	0.8006*	0.8849*	0.8632*	0.8715*	0.8445*	0.8203*	0.7487*	0.7487*	0.7424*	0.8322*	0.4864*
	(90.0093)	(118.0227)	(116.5338)	(113.4416)	(108.2861)	(103.0780)	(81.4819)	(76.2157)	(75.4766)	(113.4866)	(41.7970)
size sim.	0.0060	0.0050	0.0091*	−0.0029	0.0043	0.0065**	0.0341*	−0.0107*	0.0399*	−0.0075*	0.0611*
	(1.6615)	(1.7143)	(3.1677)	(−0.9838)	(1.4652)	(2.2596)	(10.3051)	(−2.9503)	(11.4208)	(−2.9732)	(14.5607)
cash sim.	−0.0005	−0.0008	0.0018	0.0007	0.0006	−0.0066	−0.0141*	0.0266*	0.0147*	−0.0032	−0.0114**
	(−0.1277)	(−0.2470)	(0.5428)	(0.2172)	(0.1759)	(−1.8847)	(−3.3592)	(5.5177)	(3.3009)	(−0.9889)	(−2.2019)
security sim.	0.0080	−0.0092	0.0231*	−0.0105	0.0122	−0.0251*	0.0478*	−0.0585*	0.0351*	−0.0125	0.0216
	(0.9258)	(−1.2926)	(4.0494)	(−1.8179)	(1.8843)	(−3.5514)	(5.3901)	(−6.8271)	(4.0767)	(−1.9369)	(1.9246)
total loan sim.	−0.0519**	0.1067*	−0.0533*	0.0439*	0.0124	0.0390**	−0.0596*	0.0614*	−0.0140	0.0155	−0.0728**
	(−2.1522)	(4.8998)	(−2.9795)	(2.6230)	(0.7231)	(2.0625)	(−2.7781)	(2.9073)	(−0.6128)	(1.0413)	(−2.5660)
deposit sim.	−0.0015	0.0538	0.0526**	−0.0064	−0.0058	0.0103	0.0314**	−0.0091	−0.0365**	0.0186	0.0349
	(−0.0328)	(1.1750)	(2.5021)	(−0.5809)	(−0.4580)	(0.8822)	(2.5702)	(−0.7168)	(−2.5389)	(1.8017)	(1.6695)
equity sim.	−0.0079	−0.0060	0.0064	0.0054	−0.0038	0.0116**	0.0000	−0.0033	0.0122	0.0003	−0.0038
	(−0.8756)	(−0.9364)	(1.4048)	(0.9302)	(−0.6152)	(2.3902)	(0.0056)	(−0.4633)	(1.8506)	(0.0540)	(−0.4196)
income sim.	−0.0001	−0.0000	0.0003	−0.0000	0.0000	−0.0001	0.0000	−0.0001**	−0.0000	0.0000	0.0000
	(−0.7701)	(−0.6545)	(1.3793)	(−0.3012)	(0.7529)	(−1.4078)	(0.5311)	(−2.0338)	(−0.0081)	(1.0632)	(0.2742)
location sim.	0.0803*	0.0393*	0.0040	0.0290*	0.0090	0.0510*	0.0691*	0.0700*	0.0678*	0.0563*	0.1964*
	(11.7498)	(6.9823)	(0.7629)	(5.6214)	(1.7480)	(9.5709)	(11.8085)	(11.3150)	(10.7744)	(12.0352)	(23.6627)
type sim.	−0.0015	−0.0193	−0.0096**	0.0033	−0.0009	−0.0065	−0.0206*	0.0126*	−0.0154*	−0.0005	−0.03087*
	(−0.0882)	(−1.3416)	(−1.9606)	(1.0842)	(−0.2658)	(−1.9389)	(−5.7018)	(3.3710)	(−4.1518)	(−0.2044)	(−6.5420)
observations	4032	3906	4290	4556	4422	4830	4970	4830	4692	5112	4830
adj. R-squared	0.7274	0.8224	0.7963	0.7760	0.7587	0.7277	0.6384	0.6186	0.6294	0.7690	0.4238

Notes: The dependent variable $z_{ij,Loan}^{fil.sys.}(\tau)$ is the Fisher transformation of the Pearson correlation coefficient $C_{ij,Loan}^{fil.sys.}(\tau)$. The model $z_{ij,Loan}^{fil.sys.}(\tau) = \beta_0 z_{ij,Loan}^{fil.sys.}(\tau - 1) + \sum_{k=1}^9 \beta_k \text{Similarity}_{ij}^k(\tau - 1) + \epsilon_{ij}(\tau)$ is estimated by OLS for each year τ . We report the coefficients for the lagged dependent variable as well as for the nine balance-sheet-based similarity variables in the table below with t-statistics in parentheses. * and ** indicate statistical significance at the 1% and 5% levels, respectively.

generally find that in the different market layers, the correlations can have different structures and temporal dynamics. Banks can respond differently to risk factors, and those with a high level of stock return co-movements do not necessarily synchronise their loan provisions to firms and vice versa. In addition, our results show that although there is a certain degree of persistence, the interactions among banks in each market layer also evolve, with different bank hubs dominating in different phases.

Moreover, via decomposing the total variance into different parts corresponding to different systematic and non-systematic risk components, we can further examine and shed light on how individual banks are influenced differently by these risk components. After filtering out the effects of the market mode, we further explore the determinants of non-systematic correlations by utilising the characteristics and the balance sheet information of banks. We find that this fundamental information of banks can only partially account for the residual correlations. Our conjecture for this observation is that other factors than the fundamental information and characteristics of banks may be also responsible for the emergence of co-movements in the two market layers, especially those originating from the demand side (e.g. other selection criteria of investors in the stock market, characteristics of borrower firms in the credit market). In addition, it is worth emphasising that our different models may simplify the true underlying dynamics, which may actually arise from the dynamics of higher-order correlations beyond the second order (Schweigler *et al.* 2017, Santoro *et al.* 2023, Boccaletti *et al.* 2023). Hence, what we observe and examine in this paper may just be from a simple (pairwise) angle of the more complicated multi-faceted inter-dependencies among banks.

All in all, our work contributes to the pertinent literature by providing additional evidence on the multiple potential channels of contagion and amplification of shocks among financial institutions. On top of the stock return co-movements that have often been studied in the pertinent literature (e.g. see Billio *et al.* 2012, Brunetti *et al.* 2019), our evidence suggests that loan portfolio correlations can act as an independent layer with its distinct features and that such a layer may reveal an additional source of risks in the banking system.

Our present work could be extended in several ways. First, we have focussed on the correlation layers arising from the equity and credit markets. However, it would be interesting to compare these two layers with those capturing the other important types of banks' interactions such as the physical interbank lending network and the derivatives transaction network in Japan over different periods (e.g. see Imakubo and Soejima 2010, Poledna *et al.* 2015, Brunetti *et al.* 2019, Bardoscia *et al.* 2019). Second, since the interactions considered in our present work are undirected (symmetric), future studies on the directed type of linkages between banks, e.g. Granger causality linkages among time series of stock returns would reveal further interesting patterns of banks' relations (e.g. Diebold and Yilmaz 2014, Diebold and Yilmaz 2015, Brunetti *et al.* 2019). Third, it would be interesting to extend our current analysis that only focuses on pairwise statistics into the higher-order or group inter-dependencies

among banks in the different layers (e.g. MacMahon and Garlaschelli 2015, Santoro *et al.* 2023). Such an extension can be useful to uncover the collective dynamics or higher-order effects in the banking system.

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Appendix

In this appendix, we shall present additional results for Sections 3 and 4 and provide a more detailed explanation of the method used to measure the similarity of bank characteristics, the PMFG algorithm and various network measures used to examine the extracted correlation networks.

A.1. Further decomposition of significant correlations

Following the presentation in figure 1 in the main text, we further consider different correlation parts associated with different types of the significant eigenvalues, i.e. the correlations driven by the eigenvalues exceeding $\lambda_{\max}^{RMT}$ and the correlations contributed by the eigenvalues that are below $\lambda_{\min}^{RMT}$, respectively. As illustrated in figures A1 and A2, the contributions of the significant but small eigenvalues to different pairwise correlations are negligible, whereas those of the significant but larger eigenvalues are more pronounced.

A.2. Similarity measures

For the qualitative criteria such as locations of banks, the definition of the (binary) similarity measures is straightforward:

$$Similarity_{ij}^{(location)} = \begin{cases} 1 & \text{if the headquarters of bank } i \text{ and bank } j \\ & \text{are in the same prefectural location,} \\ 0 & \text{otherwise.} \end{cases} \quad (A1)$$

In a similar vein, with different types of banks (regional banks, trust banks, city banks), we can define the measure for type similarity.

In contrast, for the quantitative criteria such as sizes, equities, deposits, amounts of cash, securities, total loans, and incomes, the degrees of similarity are not discrete but can take any value within a certain range, depending on the measurement method. Selecting the size as an example criterion, in the present work we define the similarity level between bank i and bank j in terms of their sizes as

$$Similarity_{ij}^{(size)} = 1 - \frac{|Size_i - Size_j|}{Size_i + Size_j}, \quad (A2)$$

where $Size_i$ and $Size_j$ are the sizes of bank i and bank j , respectively. Intuitively, it can be easily shown that $Similarity_{ij}^{(size)} = 1$ if and only if the two banks have the same size, and $\lim_{\frac{Size_i}{Size_j} \rightarrow 0} Similarity_{ij}^{(size)} = \lim_{\frac{Size_i}{Size_j} \rightarrow +\infty} Similarity_{ij}^{(size)} = 0$, i.e. when sizes become very different, the associated similarity goes to zero.

A.3. PMFG algorithm and network measures

Under the PMFG algorithm, first, we sort the elements of a correlation matrix in descending order. In the next step, starting with the pair of nodes i and j that have a larger degree of correlation, we add a link between them if and only if the graph obtained after the link insertion satisfies the given topological constraints. Note that although it is somewhat similar to the algorithm used to construct a MST, we allow for the presence of 3- and 4- closed cycles and motifs in the PMFG algorithm. As a result, the number of links of the inferred network is in the order of $(3N_B - 6)$ (e.g. see Mantegna 1999, Tumminello et al. 2005, Di Matteo et al. 2010, Aste et al. 2010, Massara et al. 2016, for further details).

In our study, we obtain an adjacency matrix $A_{Stock}^{PMFG} = \{A_{ij,Stock}^{PMFG}\}$ from $(C_{Stock}^{sys} + C_{Stock}^{gr.})$ and an adjacency matrix $A_{Loan}^{PMFG} = \{A_{ij,Loan}^{PMFG}\}$ from $(C_{Loan}^{sys} + C_{Loan}^{gr.})$. These, therefore, represent the binary networks inferred from the significant correlations among banks in the two different market layers. In the next step, given the adjacency matrix $A^{PMFG} = \{A_{ij}^{PMFG}\}$ (either A_{Stock}^{PMFG} or A_{Loan}^{PMFG}) for each year, we compute the following network metrics (e.g. see Newman 2010, Barabási 2016, for further details):

- The degree of the each bank i

$$k_i = \sum_{j \neq i} A_{ij}^{PMFG} \quad (A3)$$

indicates the number of other banks that have significant links with it. Based on the degrees of banks in the inferred network, we can identify banks that have more (statistically and topologically) significant correlations with other banks in each layer.

- The overall assortativity level of the network shows the type of mixing (assortative or disassortative mixing) between nodes. The assortativity level is measured by the Pearson's correlation coefficient between the degree sequences of nodes at the ends of edges (see also Luu et al. 2017). In our application to the inferred network, a negative assortativity level indicates that banks that have high degrees (low degrees) tend to link with banks that have low (high) degrees. In contrast, a positive assortativity level shows that overall banks are connected to others having a similar level of degrees.
- The average of local clustering coefficients is defined as

$$\bar{c} = \frac{1}{N_B} \sum_{i=1}^{N_B} c_i, \quad (A4)$$

where each c_i is the local clustering coefficient of bank i (see also Luu et al. 2017). Given the adjacency matrix $A^{PMFG} = \{A_{ij}^{PMFG}\}$ of the inferred network, c_i is defined as

$$c_i = \frac{\sum_{j \neq i} \sum_{k \neq i,j} A_{ij}^{PMFG} A_{jk}^{PMFG} A_{ki}^{PMFG}}{\sum_{j \neq i} \sum_{k \neq i,j} A_{ij}^{PMFG} A_{ik}^{PMFG}}. \quad (A5)$$

Mathematically, each c_i indicates the probability of the 'neighbours' of node i that are also connected to each other, thus, showing the clustering behaviour around node i (as the centre).

- The average shortest path is the average of the shortest distances between any two different nodes in the network

$$\bar{l} = \frac{1}{N_B(N_B - 1)} \sum_{i \neq j} d_{ij}, \quad (A6)$$

where d_{ij} is the shortest length of any path between i and j .

We summarise the evolution of the different metrics for the inferred networks of the stock return correlations and loan portfolio correlations in figures A6 to A13. As we can see, although the density is constrained,[†] the two PMFG networks exhibit different topological characteristics and different levels of clusterings, assortativity levels, and shortest distance averages among banks over the years.

[†] The output of applying PMFG method to a correlation matrix is a connected planar graph with about $(3N_B - 6)$ edges. Hence, the densities of the two filtered graphs should be identical or similar.

A.4. Additional results

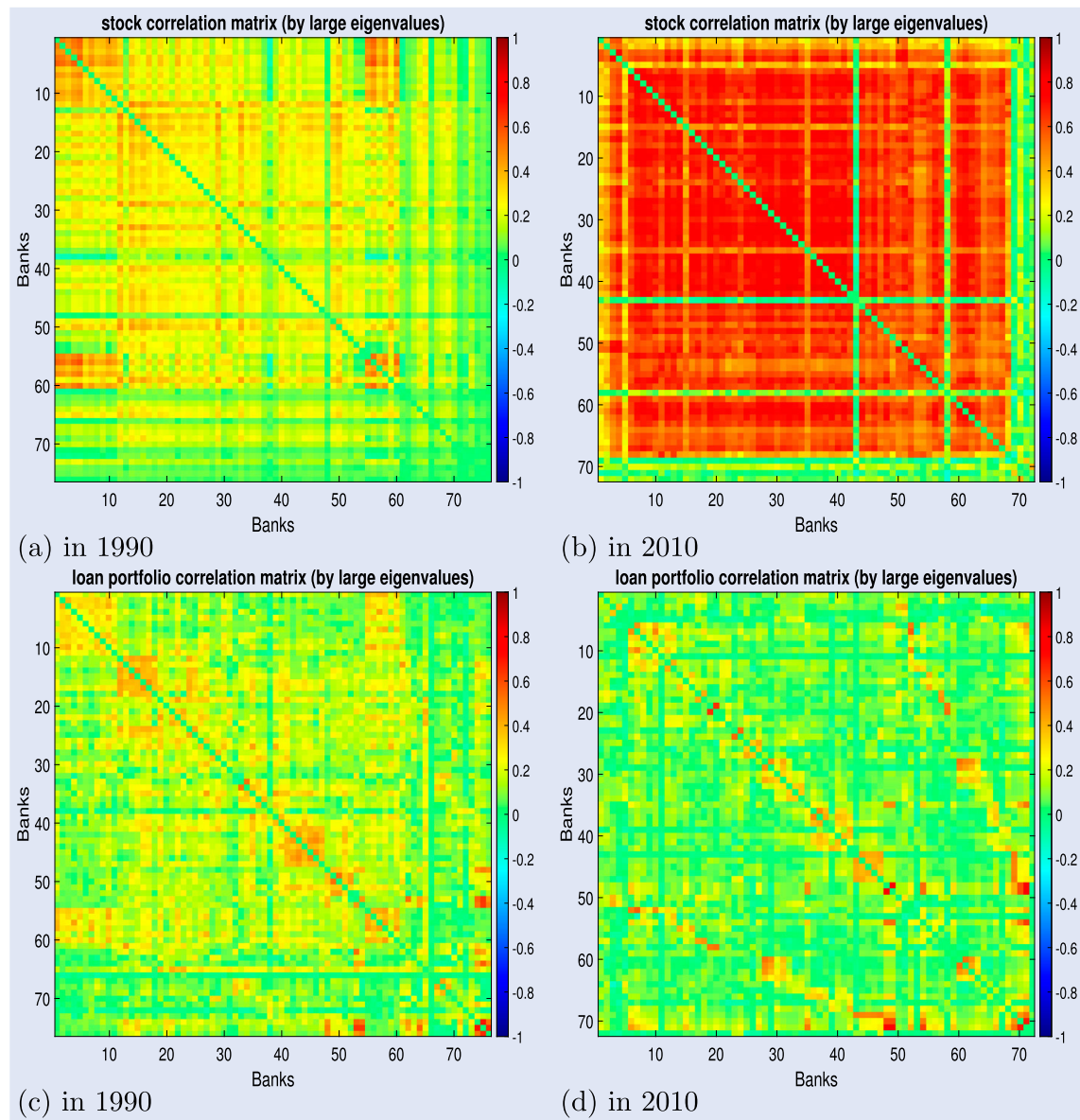


Figure A1. Different layers of correlations among Japanese banks contributed by the large eigenvalues that are beyond $\lambda_{\max}^{RMT}$, in 1990 and 2010 as examples. The colorbar indicates the correlation levels ranging from -1 to 1 . (a) in 1990. (b) in 2010. (c) in 1990 and (d) in 2010.

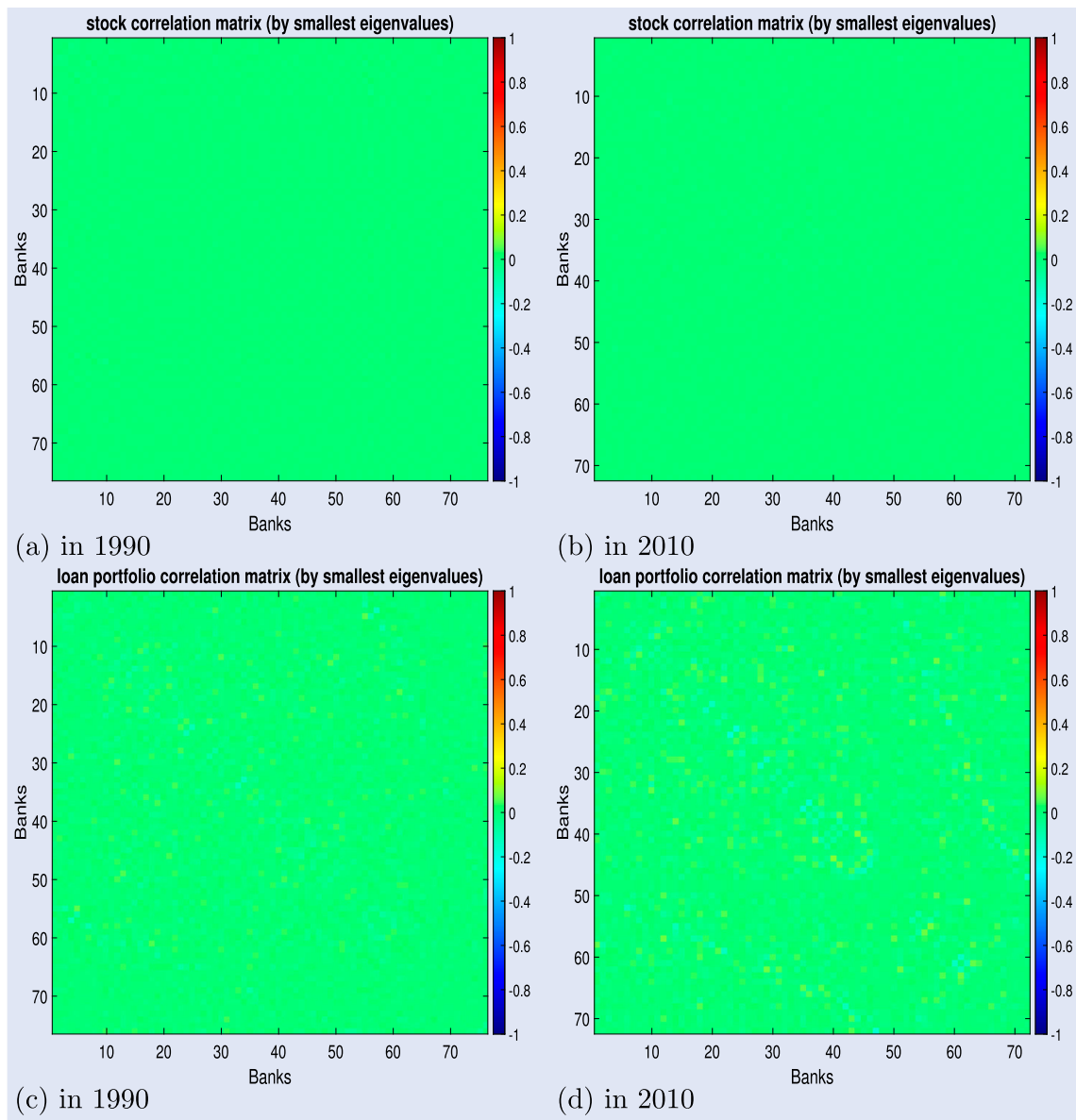


Figure A2. Different layers of correlations among Japanese banks contributed by the eigenvalues that are smaller than $\lambda_{\min}^{RMT}$, in 1990 and 2010 as examples. The colorbar indicates the correlation levels ranging from -1 to 1 . (a) in 1990. (b) in 2010. (c) in 1990 and (d) in 2010.

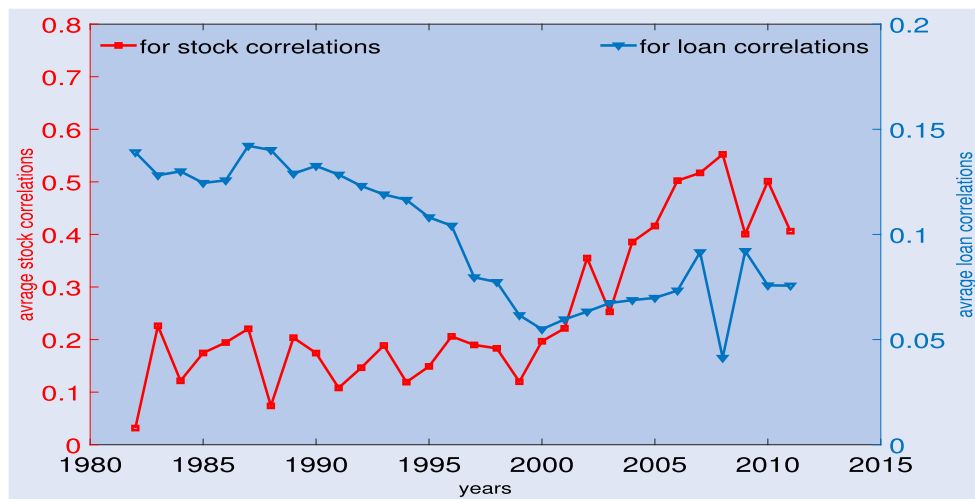


Figure A3. Correlation averages in the two layers over the period from 1982 to 2011.

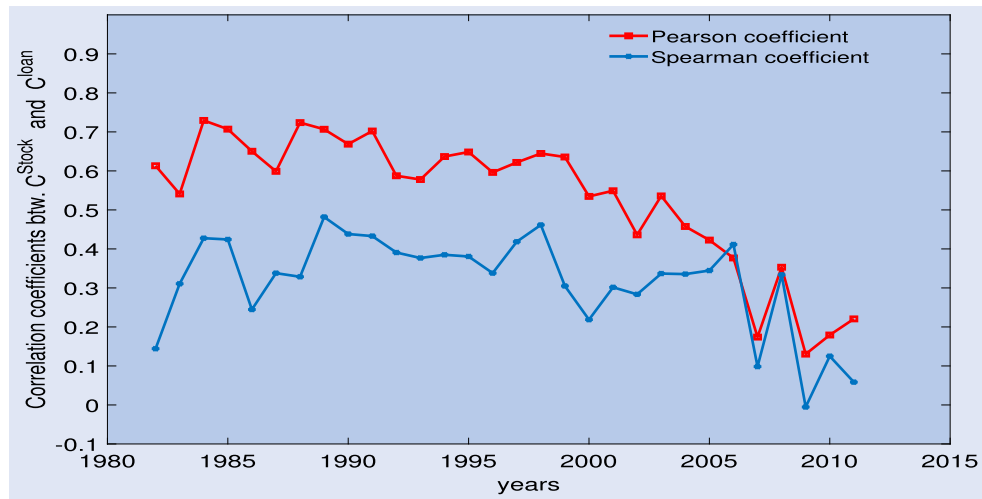


Figure A4. Pearson and Spearman correlation coefficients between $\{C_{ij}^{Stock}\}$ and $\{C_{ij}^{Loan}\}$ over the period from 1982 to 2011.

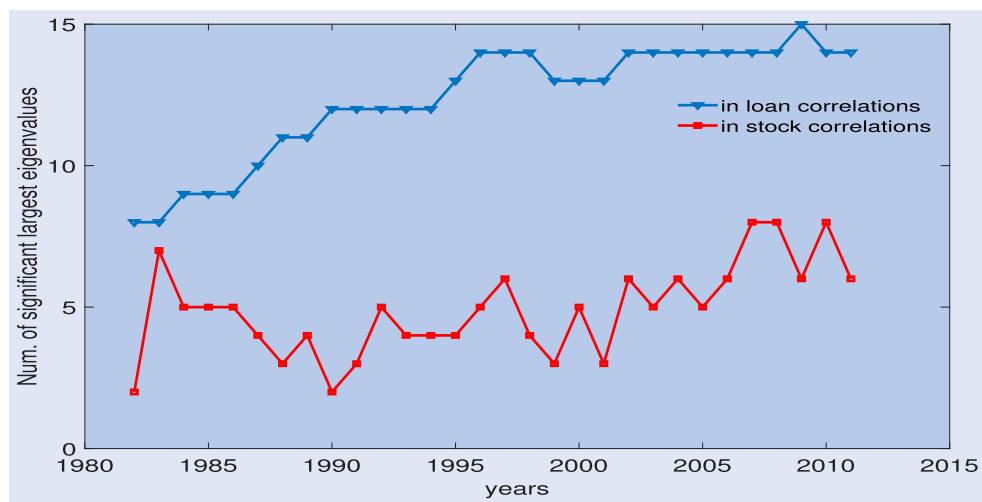


Figure A5. The numbers of significant eigenvalues in the two correlation layers over the years.

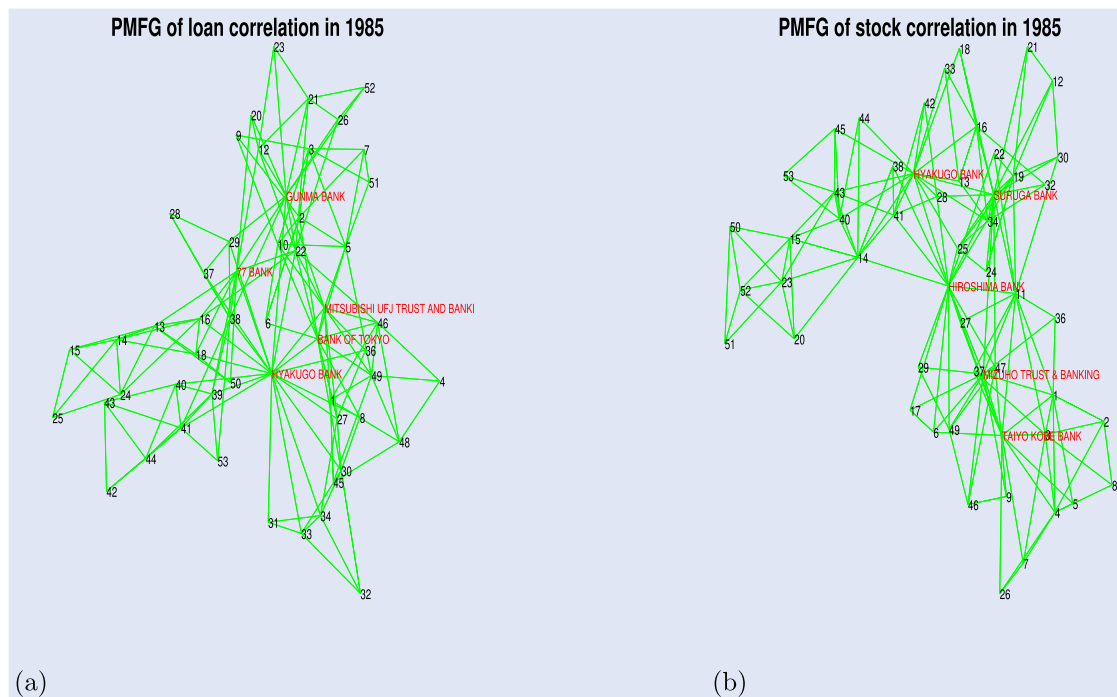
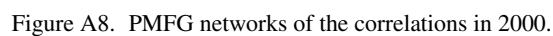
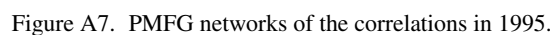


Figure A6. PMFG networks of the correlations in 1985.



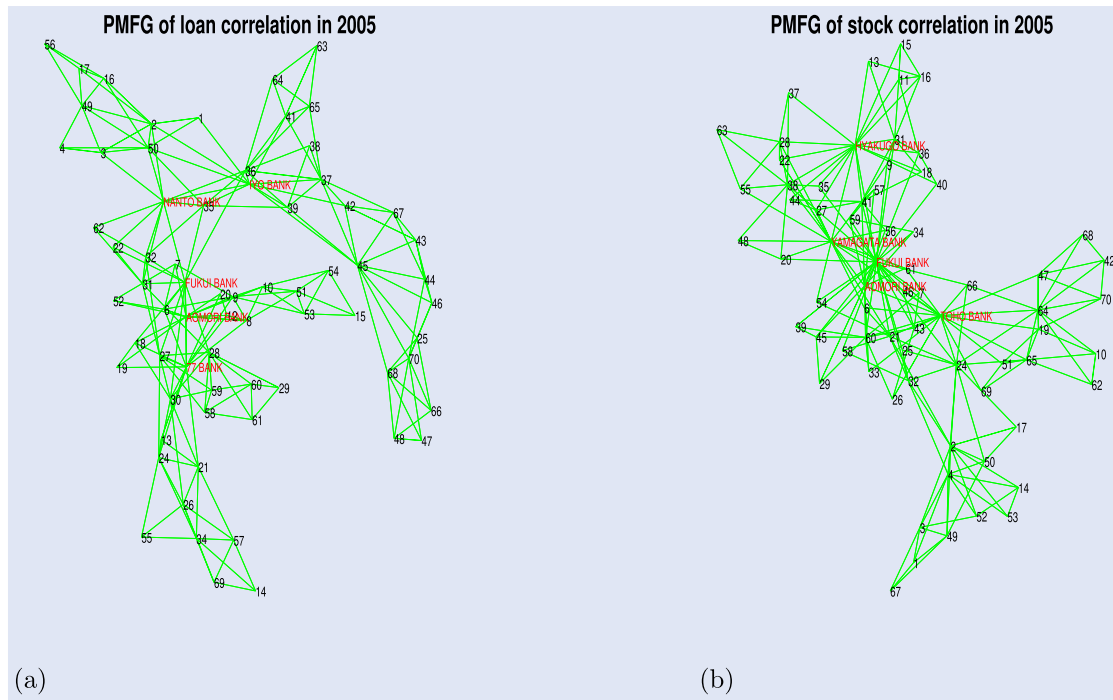


Figure A9. PMFG networks of the correlations in 2005.

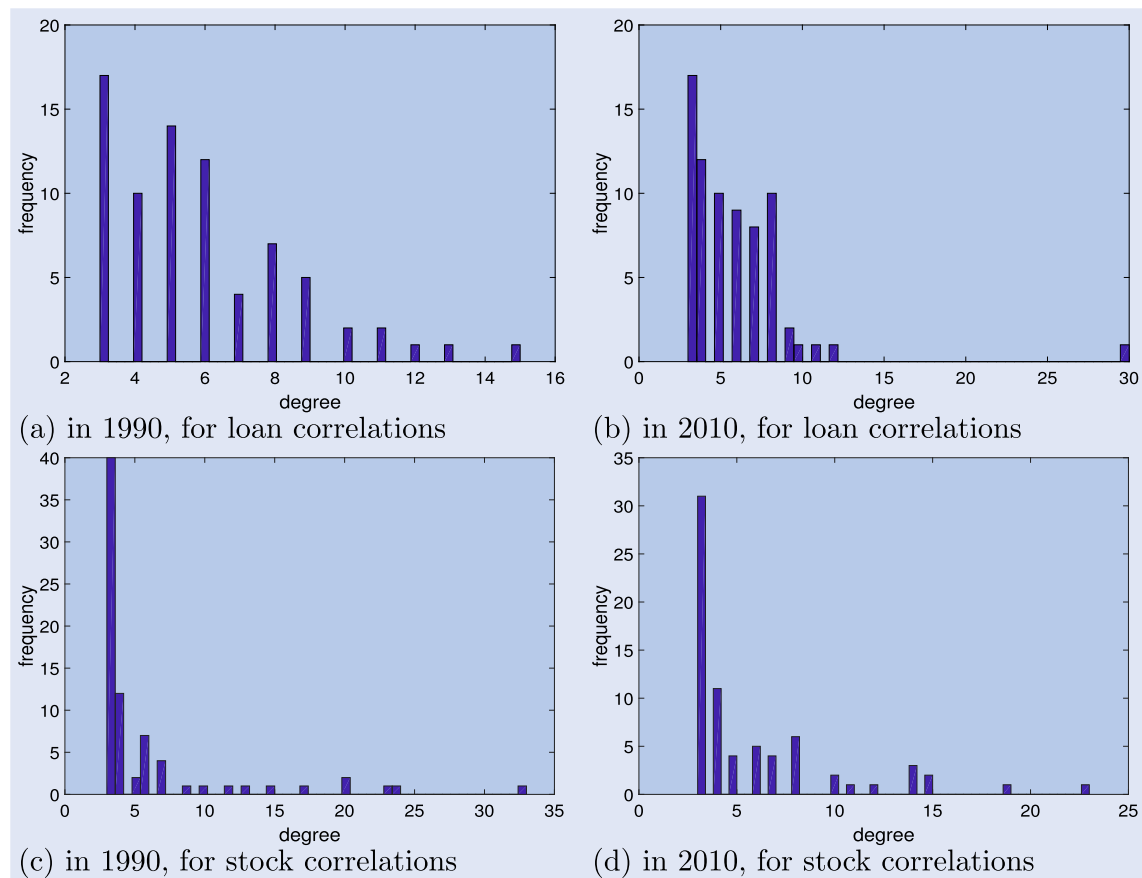


Figure A10. Degree distributions of the PMFG networks inferred from correlation matrices, in 1990 and 2010 as examples. (a) in 1990, for loan correlations. (b) in 2010, for loan correlations. (c) in 1990, for stock correlations and (d) in 2010, for stock correlations.

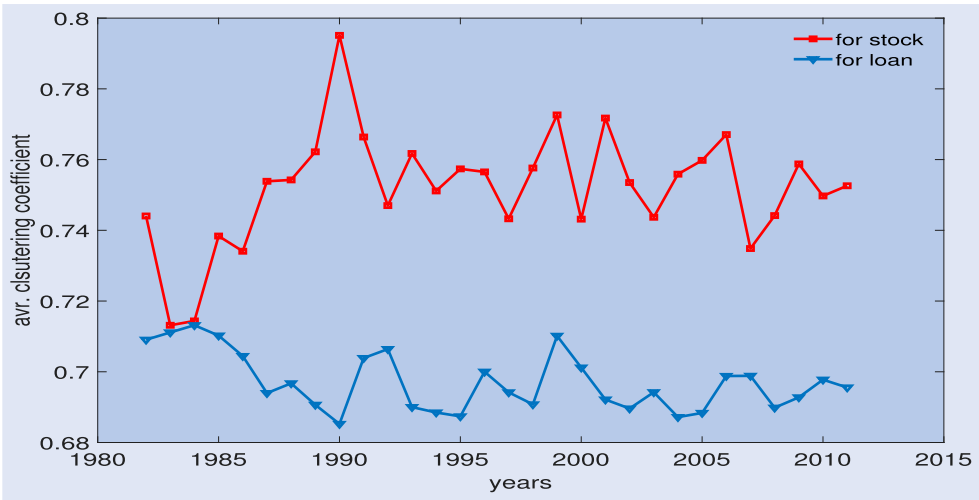


Figure A11. The average of local clustering coefficients of the inferred networks over the years.

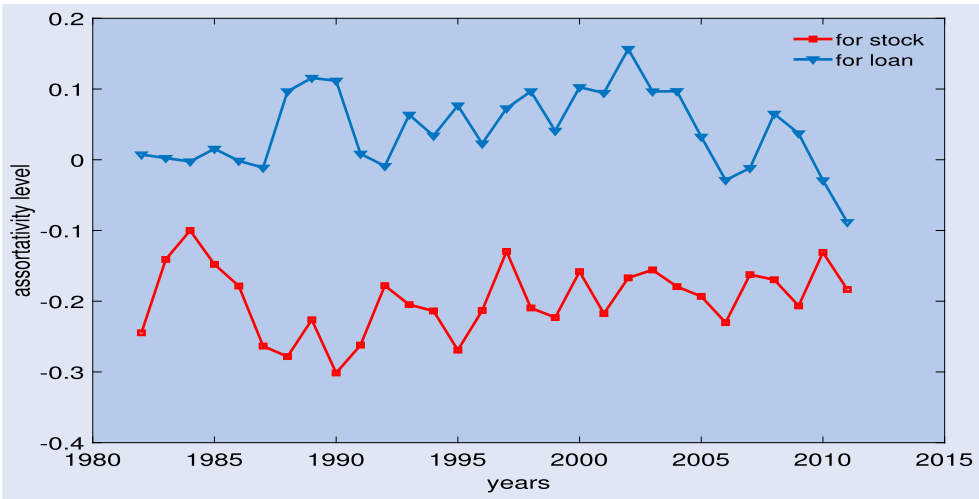


Figure A12. The degree of assortativity of the inferred network in each layer over the years.

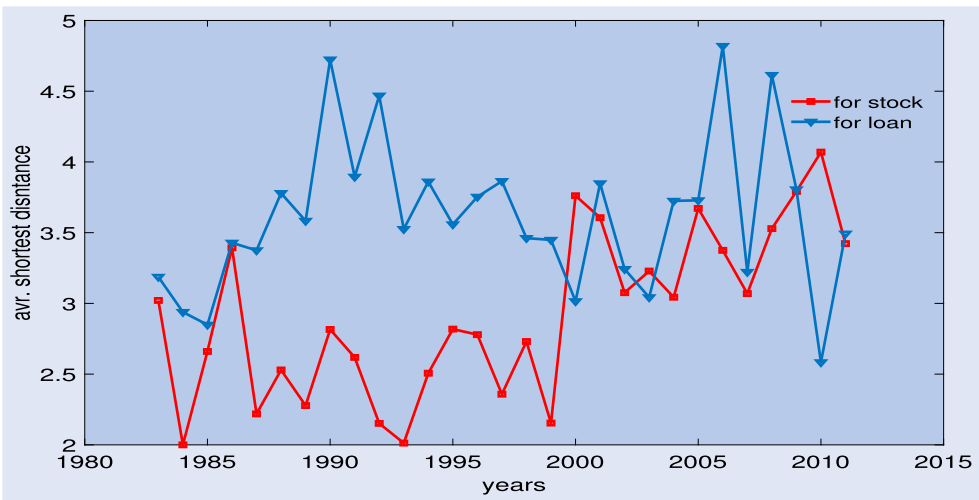


Figure A13. The average shortest distance of the inferred network in each layer over the years.